



MSc in Computer Games Technology

Project Report

2025

Second Person – Exploring game worlds from a new perspective

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November 25, 2025

Declaration

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Abstract

This report documents the research and process of exploring video game worlds from the second person perspective. The unity engine was used to create a multitude of working prototypes that positions the camera in a way that showcases the second person within a playable media. Genres of each prototype explore Puzzle, Racing, Shooter and Adventure displaying the possibilities of what can be achieved within the second person. Playtest sessions were hosted to allow participants to provide feedback and thoughts by exploring the familiar genres from a perspective that they've never experienced in mainstream games.

Keywords: Second Person Perspective (2PP), Camera, Video Game, Alternate Viewpoint, Separation, Observed Embodiment

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1 Introduction

Games throughout the years use the first- and third-person perspective to tell their story and to drive engagement for players, however what's often disregarded is the perspective that's rarely used in today's games: the second person perspective. Within this report illustrates the process of designing and building 4 playable mini prototypes using the second person perspective and researching how it can influence immersion and what gameplay opportunities could be offered compared to other perspectives used in video games. To best describe and explore the unique perspective within games introduces a conceptual term of **Observed Embodiment**.

What is Observed Embodiment? Observed Embodiment is a term created to label the technique and features used to apply within games that use the second person perspective. The term defines the second person gameplay as an act of *being seen* rather than *seeing*. The player controls a character whose actions and identity are observed through the viewpoint of another entity. In short, it's seeing yourself the player from the perspective of another person within the game world.

Observed – ‘notice or perceive (something) and register it as being significant.’

Embodiment – ‘a tangible or visible form of an idea, quality, or feeling.’

Throughout the course of this project, the creation of multiple prototypes will showcase the how the second person perspective can be utilised within different genres of games is documented. These prototypes were used to conduct playtests where users interacted with the game to gather feedback and information based on their experiences. Within this introductory chapter explains the discovered problem, motivations behind the project, beneficiaries for those interested in the project and objectives considered. Since the term ‘Second Person Perspective’ will be used frequently throughout the project, it will be referred as ‘2PP’.



Figure 1 - First-, Second- & Third-Person Perspective in Games (Robinson, 2019)

1.1 The Problem

Many video games tell the player's story in the game through the protagonist's eyes and perspective either through the first- or third-person perspective (FPP & TPP). According to the Steam Database, 3564 first-person (SteamDB, 2025a) and 2085 third person (SteamDB, 2025b) games were released in 2024, highlighting the dominance of these perspectives and implying that there is a gap for providing an experience in an alternative perspective. The other alternative perspective that could be utilised within games is by telling a story or providing an experience within the second person perspective.

Why specifically is the second person the primary focus for this project? The reason being is that this perspective is rarely used within games, remaining underexplored and restricted to short, experimental gameplay experiences or in text-based adventure games. Examples to note that visually demonstrate and execute the second person well are; *Driver: San Francisco* in which the player controls their own car from the point of view of the pursuing vehicle and *Psychonauts*, which features a boss called The Den Mother, where the stage turns pitch black, hindering the player's vision and uses the boss' infrared vision to see themselves.

A brief definition of the second person perspective in video games would be “within a 2nd-person game, the player would see the character they're controlling through the perspective of another character interacting with them making you the character.” (Chollet, 2022). Something this simple could open a wide possibility of gameplay and story experiences that first and third person games may not be able to provide because of the uniqueness of the second person.

This leads to the problem faced with the existence of first- and third-person games providing a fun, thrilling gameplay experience debates whether the second-person perspective could deliver a comparable experience. In games like *Driver: San Francisco* and *Psychonauts* restricted the perspective in a short, experimental sequence could've been due to technical and narrative challenges that best fit the gameplay and story in those short sequences.

With second person, instead of seeing the game world from the character's perspective, the player would control the character but see themselves from the perspective of another person. This would be the foundation of Observed Embodiment by investigating how this perspective could influence immersions compared to other perspectives.

1.2 Motivation

One of the main reasons for undertaking this project is to be given the opportunity to explore the idea behind the second person perspective and see what kind of creative visions that could be achieved with this point of view. As someone who enjoys making and playing games, this project will test the knowledge, skillset and creative ability to produce a product that may serve as a base foundation for those willing to create something related to the second person.

Since many people don't know about the second person in games, it would be interesting to view how others would perceive the idea of playing a game through the eyes of another person yet still having the familiarity of controls and gameplay styles that most people are familiar with, for example a puzzle game, racing game or a shooter game. This would offer immersive and more engaging gameplay experiences that would be unique to the point of view genres.

Through designing and creating a prototype, the aim is to contribute to the discussion around the alternative perspective in games and to highlight the creative potential within this underexplored subject, hoping to inspire future designers and developers to explore, interact and experiment this point of view to create something entirely new for people around the world to experience.

1.3 Beneficiaries

The project & documentation will benefit a range of interest groups within game development and academic research.

Table 1 - Beneficiaries

Beneficiaries	
Group	Benefits
Game Developers & Narrative Designers	Will gain insight into how the second person perspective could be utilised as a main perspective within a game to enhance a player's experience, immersion and engagement. By producing a playable prototype that showcases the uses of 2PP, will offer designers an interest in expanding and enhancing gameplay elements with the perspective.
Academics & Researchers	Those that explore perspectives within literature may find this project beneficial as it could lead to the expansion of a perspective that's mainly used in fictional literature.
Gamers & Gaming Community	They may benefit from the greater exposure of an alternative narrative experience – such as someone who is familiar with first and third person games. This project can serve as an introduction to a less common but interesting way of exploring and interacting within game worlds which could bring more exposure to the perspective genre.
Author	The most important and main beneficiary of the project who will garner experience in both game development and narrative design with the use of the second person perspective as the experience gained will be from learning, developing & prototyping a game for the purpose of this project.

1.4 Objectives

The objectives for this project, along with the outcomes that are intended to be achieved are described below.

1. **To analyse the second person perspective within existing video games and interactive narratives** – Explore existing examples as to where the second person perspective has been used (e.g. Driver: San Francisco, Psychonauts) to understand how it has been applied and implemented in different genres. This would aid in creating a foundation for experimenting with the perspective.
2. **Explore how the second person perspective impacts the player's immersion** – By creating a prototype with different scenarios in unity to evaluate the player's immersion to determine if the second person enhances or disrupts immersion.
3. **To identify the potential gameplay experiences that are unique to the second person perspective** – Investigating how the perspective could create novel mechanics and new ways to engage within game worlds that wouldn't be possible in first or third person.
4. **Compare the second person perspective with first and third person perspectives in relation to storytelling** – To identify how the second person perspective affects narrative delivery, player identification and emotional engagement as opposed to other perspectives.
5. **Evaluate player's perceptions of the second person perspective through user testing** – By conducting play test sessions using the IEQ to collective qualitative data on how the player's experience the second person and by calculating the challenge, control, real world dissociation, emotional and cognitive involvement.

1.5 Organisation of the Report

This report is structured into 5 main chapters:

- Chapter 2 – Critical Context: Details the existence of second person in games and media.
- Chapter 3 – Methods: Details the methods of how each playable prototype were created.
- Chapter 4 – Results: Details the results recorded from conducted playtest sessions.
- Chapter 5 – Discussion: Details the discussion of each objective.
- Chapter 6 – Evaluation, Reflection & Conclusion.

2 Critical Context

2.1 Introduction

The second person perspective within games remains underexplored as opposed to the more common and used first and third person formats. Often utilised within shorter sequences or in text-based games (e.g. The Sumerian Game, 1964), the second person has rarely been considered for a full gameplay experience. This critical context will examine the existing implementation of the second person in existing games, the comparison between first and third person perspectives, the definition and use of the second person in other mediums such as literature and finally immersion in games.

2.2 First- and Third-Person Perspective

When creating games, choosing the perspective of the player is one of the most important design choices. Common camera selections that can be chosen in some games include isometric, side scroller, target camera and for perspectives either first or third person points of view. It could be suggested that creating a game that utilises different viewpoints through which the player sees can create different experiences, as an example the First person POV perceives the game world through the eyes of the character, being able to observe the world around them up close and gives them a clearer view of the scenery that's in front of them. The first-person perspective is mostly believed to be the most immersive feeling experience for the player. The third-person perspective observes the main character in frame of the scene, without giving the player the sense that they are the character.

By looking at Denisova and Cairns' publication of "First Person vs. Third Person Perspective in Digital Games: Do Player Preferences Affect Immersion?", the aim of their study was to research whether interacting with a game in the first-person perspective is more immersive than playing in the third person point of view. The procedure for their experiment, participants played 15 minutes of a quest in a game and after the time was up, they'd fill out the IEQ (Immersive Experience Questionnaire) and were asked about their preferred perspectives. Results in their experiment shows that participants who played the game in the first-person perspective had higher levels of immersion than those who experienced the game in the third person (watching their character from behind) (Denisova and Cairns, 2015).

Using a similar method to Denisova and Cairns, in this research study, participants will play a selection of scenarios built using the second person perspective whilst providing insightful feedback and using a selection of questions based off the IEQ for them to fill out after playing each scenario. Each scenario will have different levels of immersion scored.

2.3 The Second Person Perspective

When discussing the differences and uses of perspectives, it comes from the conjugation of verbs and the related pronouns. Using the correct pronoun within a given context is suggestive as it can be applied to the first-, second- and third-person singular pronouns to epistemic situations. With a couple of examples, when talking about different perspectives: “*I* have a headache” applies to the first-person perspective, “*you* have a headache” to the second-person perspective and “*he/she/it* has a headache” applies to the third person perspective. Given the perspectives are related between epistemic subjects and objects, it can be denoted that referring to the *first*-person means being related to oneself, the second or third person refers to ‘someone’ (“you”, “he”, “she”) or something (“it”) else. What is the difference between the second- and third-person perspective? In some part, one difference is referring someone as “he”, “she”, or “it” in one case and “you” on the other. More precisely, it’s the difference between a perspective on some object, whatever the object may be, and an “object” that’s recognised as an epistemic subject herself, that is, recognised as someone who has a perspective of her own and is a possible partner of an interactive process. (Pauen, 2012)

While the use of the second person perspective within games is uncommon, the use of the perspective is widely used within literature and media. Within literature, Second-person narration is a little-used technique of narrative in which the action is driven by a character ascribed to the reader, one known as *you*. The reader is immersed into the narrative as a character involved in the story. The narrator describes what “you” do and lets you into your own thoughts and background. The most well-known piece of fiction that employs second-person narration might be Jay McInerney’s novel *Bright Lights, Big City* (Merriam-Webster, n.d.). Within media, second person point of view uses the pronoun “you” to address the reader. This narrative voice implies that the reader is either the protagonist or a character in the story and the events are happening to them. The other points of view in writing are first person and third person, which includes third person omniscient, and third person limited. (MasterClass, 2021b).

2.4 Second Person in Games

In the context of video games, popular youtuber Nick Robinson documented what a “second person” video game would look like. At first, he documents that most 3D video games can be placed into one of two categories: first person where you see the game world through the actual eyes of the character and third person where you see the player character from the outside – often behind their back. He goes on to ask the question if we already know what a first-person game and a third person game looks like, then what exactly would a second-person game look like? Nick goes on to explain the meaning behind each perspective and claims that the second person is less common in narrative writing and more likely to encounter it in things such as: lists of instructions or a choose-your-own-adventure book. The analogous video game camera perspectives for first- and third-person writing are obvious, knowing what an “I” game and a “he” game

looks like, but what does a “you” game look like? Nick found the answer whilst playing a level in Ubisoft’s *Driver: San Francisco* where the task is to chase the main character (yourself) however the camera is positioned and situated inside a vehicle behind the main character’s car. He describes the mission as a chase mission; however, you’re the one being followed while simultaneously seeing it all unfold from the perspective of the car doing the following. All the elements combined makes something odd to play, and there’s something shocking and disorienting about seeing a first-person perspective on your screen but also not controlling that perspective while remotely operating the car that you’re tailing. (Robinson, 2019)

Like Denisova and Cairns’ look at the immersion in first and third person games, another publication is a research paper written by Jin Zhou from Northeastern University in Boston called “Where is my head? Explorations of Second Person Perspectives used in VR Games” which studies the second person perspective in Virtual Reality and how it can be utilized to improve player experiences. They designed a game playable in both VR and Desktop where the player’s view and whom they controlled were separated and ran an experiment to compare the difference in player’s experiences between both versions of the game. Their study involved 28 participants who played either the VR or Desktop version of their game and provided feedback through a think-aloud protocol and surveys. Their evaluation focused on 6 dimensions of player experience; meaning, curiosity, immersion, feedback, and challenge. Findings showed that the VR version of the game outperformed the desktop version in terms of immersion and meaning with a score of 6.49 for immersion when played in VR and 5.49 when played on desktop. Their research generalised the second person narrative into game design and defined second person perspective in games (Zhou, 2022).

By identifying the existing usage of the second person, applying something that’s better executed in literature will be challenging to implement in a game, but the principle will be the same. Since the second person is normally directed towards someone or the player in this case, the idea would be to develop a “You” game where the player (you) is viewed from the perspective of another being, applying the narrative of “*you* are driving a car” or “*you* are fighting the enemy”.

2.5 Narrative & Visual POV

Since the second person perspective is the primary focus for this project, finding games that already utilise the second person perspective is challenging, however there are signs to identify what makes a game second person. The ‘extreme’ use case of the second person used in games are the ones mentioned earlier such as *Driver: San Francisco* and *Psychonauts*, one game that is primarily in the second person is a game titled “Second Person Shooter Zato”. The game uses a second-person perspective to display the game from the viewpoint of the enemies looking at the player, rather than the other way around; it makes use of a split screen to show the perspectives of multiple enemies. The game’s perspective was inspired by surveillance cameras; the title takes its name from Zatoichi (a fictional, blind Japanese swordsman), due to the player

character's inability to see (Bomb, 2011). Screencheat is an interesting example which one could argue is in the second person. Screencheat is a 1-4 player arena shooter where each player can see the viewpoint of the other players. Unlike other multiplayer FPS games, player models are not rendered in game so the only way to figure out where opponents are is to look at their viewpoint on your screen, or "screen look" as it is most known (Marino, 2019). Although you may argue that the game is played in the first person, the gimmick of having to look at the opponent's screen, the argument for the game being second person would be because yourself and the opponents are invisible, you must use your opponent's POV to try and identify their position. With the theme of the second person perspective, identifying that the player views themselves through the perspective of another character in second person perspective will be known as second person in a visual point of view in games.

We know that the second person in a visual point of view is seeing the player from another character's perspective in games, the second person narrative point of view in games is slightly different. But first, what is narrative point of view in general? Point of view is the "eye" or narrative voice through which you tell a story. When you write a story, you must decide who is telling the story, and to whom they are telling it. The story could be told by a character who is involved in the story, or from a perspective that sees and knows all the characters but is not one of them (MasterClass, 2021a). a mission in the game Call of Duty: Black ops Cold War demonstrates the use of the second person narrative well. In the mission Break on Through, the story takes the player into a flashback mission in Vietnam with the main antagonist Russel Adler telling the player what you've experienced in Vietnam. Although the gameplay is displayed in first person, typical in Call of Duty, this mission is technically played in the second person narratively, not visually. What is meant by narrative second person is at the start of the mission, Adler says "*You* ran forward and picked up an M16." "...*You* set off to locate the bunker". With him saying "you" in this mission rather than "Bell" makes the gameplay in second person as opposed to first person, even though the gameplay is displayed through the eyes of the protagonist (DraKulis, 2020). Addressing the player and identifying them as "you" within the story or as a voice over regardless of which point of view the game is played at will be known as second person in a narrative point of view in games.

3 Method

This methods within this project were developed in 5 phases.

- Phase 0 – Initial Research & Project Setup
 - Phase 0 prepared for the initial start of the project. Firstly, existing methods and implementations of the second person perspective were reviewed and cited in the Critical Context section. Next setting up the Unity project such as finding suitable assets and integrating them into the engine were key for quick prototyping.
- Phase 1 – Creation & Design of 2PP Features and Gameplay
 - Phase 1 consisted of sketches such as level design and gameplay elements using pen and paper to quickly sketch ideas and layouts to observe how the different perspectives may play out.
- Phase 2 – Implementation of Phase 1
 - Phase 2 implemented the sketched ideas from Phase 1 into Unity and online assets were used to develop and create the prototypes showcasing the second person, using the sketches as a baseline.
- Phase 3 – Prototype Playtests for User Evaluation
 - Phase 3 used the prototypes developed in Phase 2 to be play tested to gather user feedback. The playtests helped gather an understanding on what the strengths and weakness of the second person perspective and would then influence the development of the final prototype.
- Phase 4 – Final Prototype & Play test
 - The final prototype was developed after the play test from phase 3 and learning what worked from the initial prototypes. This prototype was then tested for more user evaluation to gather more insight on the perspective.

3.1 Phase 1 - Initial Sketches

3.1.1 Idea 1: Headless Player

The first idea sketched out after researching gameplay possibilities in the second person is a Headless player idea. As shown in **figure 2** the player sees through the eyes of the head however, they cannot move the head, only the body can move. The idea for this would be a puzzle game and some puzzle elements would include a maze for the player to navigate or having to enter a room blind but see themselves on a CCTV camera.

For controlling the player within the game, as the head will be stationary and is not meant to move, the mouse would be used to look around, and the player can be controlled using W & S to move forward and backwards and A & D to turn the body left and right.

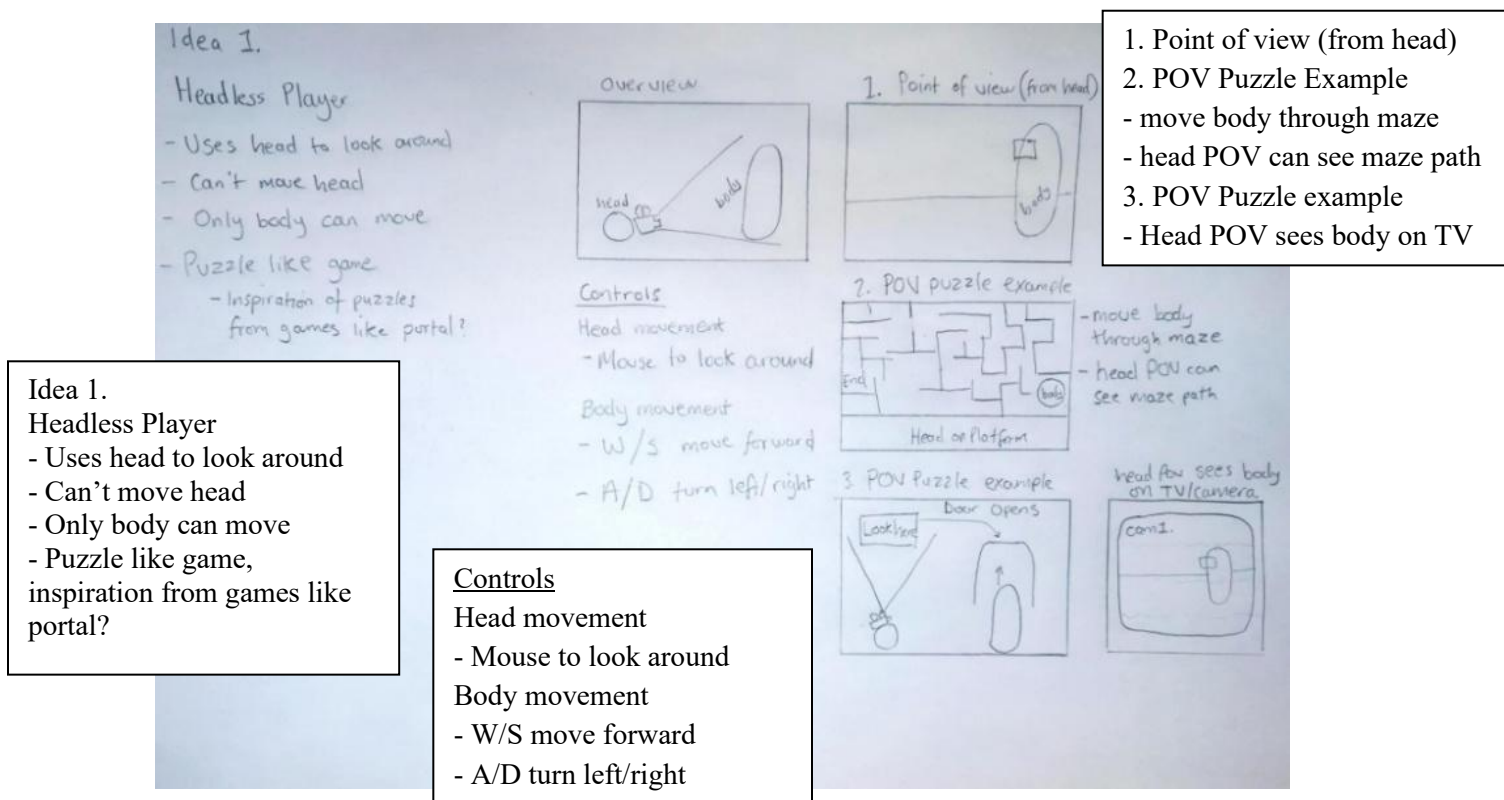


Figure 2 - Initial Sketch Idea 1: Headless Player

A few puzzle ideas came to mind when drawing the initial sketch for the headless player. One puzzle idea consists of the player being trapped in a maze, the head would be somewhere in the game world that can easily view the maze from a top-down view, being able to see and navigate the player's body through the maze. The second puzzle idea would have the player open a door and enter a room, but since the head can't see inside the room, they'd have to use a CCTV camera that's inside the room to see themselves and complete a puzzle of some sort. The last puzzle idea would be physically moving the head into places where the body won't be able to go through, and the player would be able to pick up the head to move it around.

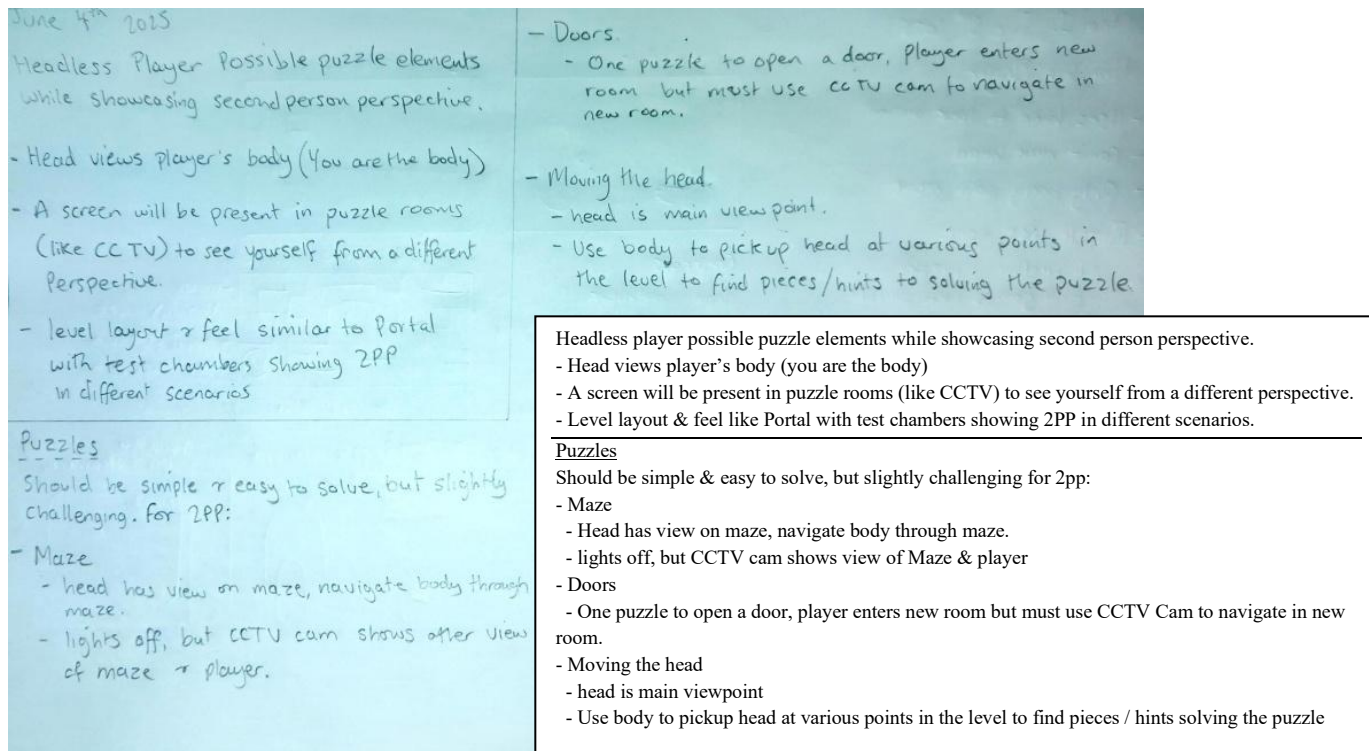


Figure 3 - Initial Sketch Idea 1: Headless Player (Back of Sheet)

3.1.2 Idea 2: Rally Driver Instructor

The second sketched idea that takes inspiration from Driver: San Francisco is a Rally Driver Instructor. The idea behind this is the camera is positioned in the passenger seat of a rally car; however, the player controls the driver that's driving the car. The objective for this idea would be the player to try and set the fastest lap time on different racecourses.

In **figure 4**, looking at the overview of the sketch, the passenger viewpoint would feature a map of the race track for the player to follow and like most rally car racers, the passenger speaks to the driver to indicate a turn or a bend coming up, for example: "half right, hard left" etc but ideally to make the scene feel more appropriate to the second person, the passenger would say "you should slow down for this left bend", "you're making good progress", "you're halfway done".

For the controls, as the player's actual viewpoint is the passenger, the mouse would be used to look around the interior of the vehicle, but the control for the actual driver is like the headless person idea where W & S is to drive forward and backwards and A & D is used to turn left and right.

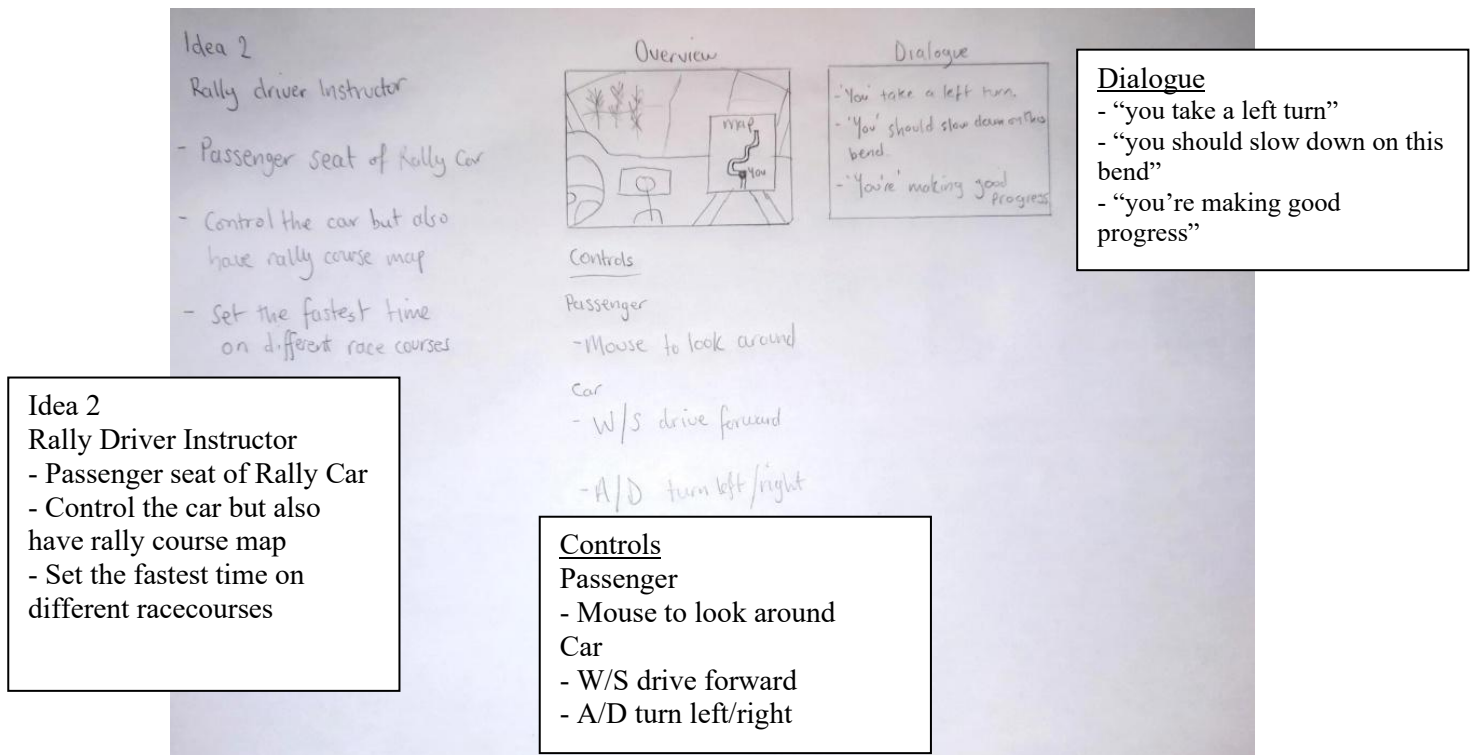


Figure 4 - Initial Sketch Idea 2: Rally Driver Instructor

3.1.3 Idea 3: Boss Rush

Lastly, **figure 5** shows the third idea inspired by the boss level from Psychonauts would be a Boss Rush level. The gameplay loop would be like Psychonauts where the player looks at themselves on the battle arena from the perspective of the boss and the idea is to fight against different bosses within the arena and exploit their weaknesses to defeat them.

When designing and sketching out ideas for different bosses with weaknesses, two ideas came to mind, one example would be a fire boss where the player would step on some buttons to make a platform fall into water hurting the boss and the other idea would be a water boss where the battle arena is surrounded by lava with broken platforms, the idea would be for the player to destroy the platform where the boss is standing to damage it.

Controlling the player would be like the headless player idea by moving the player with W & S and turning with A & D, however for the player to attack back, the right mouse button would be used to aim at the boss, and the left mouse button would shoot the player's weapon.

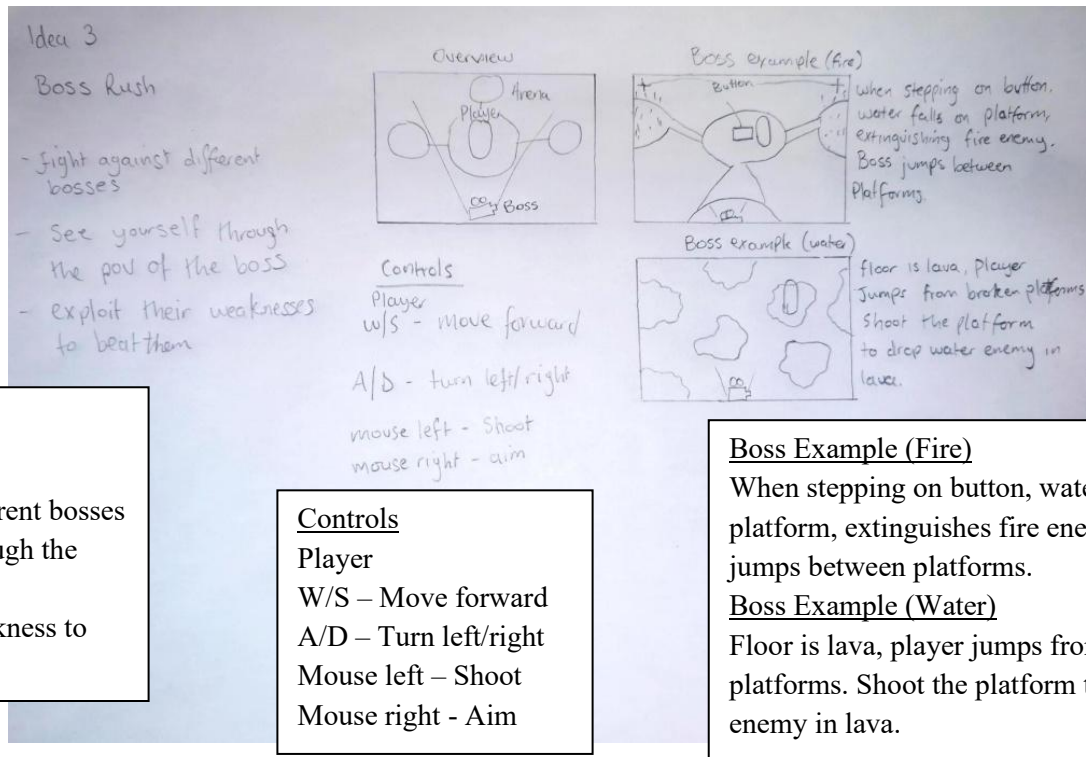


Figure 5 - Initial Sketch Idea 3: Boss Rush

3.2 Phase 2 - Sketches into Working Prototypes

After drawing and sketching out some initial ideas regarding the second person perspective in different genres of games, these sketches became the backbone of influencing the design and prototype decisions made throughout the course of the project. This section will break down the thought process and choices made and how each sketch has been digitalised into a playable and working prototype.

3.2.1 Feasibility Study

This feasibility study outlines the tools used throughout the project to accelerate the process to produce the prototypes within the allocated time for this project. The decisions made for the development of this project were within agreement of the supervisor. Below explains the game engine, assets and the use of generative AI.

Game Engine

The project prototypes were developed using the latest version of the Unity game engine, 6.0. Because of the prior experience and familiarity with using Unity in the past, this ensures that each prototype would be developed efficiently within the time frame.

Assets

As each prototype is more to showcase the uses and implementations of the second person, using previously purchased as well as free assets from the Unity Asset store allowed the development to primarily focus on the gameplay and technical aspect of the project.

GitHub Copilot AI

With some confidence in programming as well as the latest integration of Generative AI within Visual Studio, GitHub Copilot AI was used for assistance with writing out scripts and fixing errors. While majority of the implementation was done independently, the use of AI was used within most of the programming and not within the creative design of the prototypes. Prompts and code generated by AI will be under [Appendix 1](#).

3.2.2 Maze Prototype

By designing each prototype in the order of sketch designs, the ‘Headless Player’ prototype was created first. As the main viewpoint of 2PP is seeing yourself from another perspective, the design philosophy behind the first prototype was to navigate through the maze by looking through security cameras with different angles that the player could switch from for better visibility.

Using an online maze generator to create the base layout, cubes were used to plot out the walls of the maze for the player to move through. Next was figuring out the best placements for the cameras to be positioned for optimal viewing for the player to see themselves through. In the end, 4 cameras were placed above each of the 4 corners of the maze which look towards the middle of the scene.

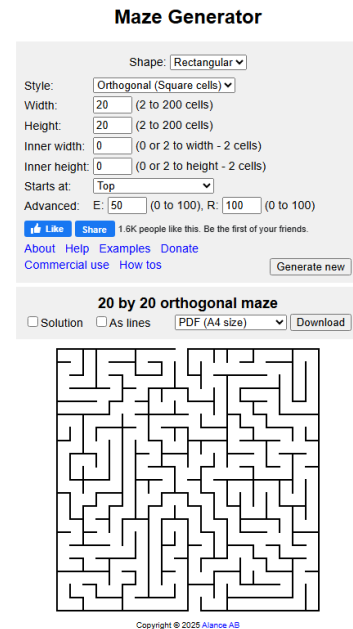


Figure 6 - Online Maze Generator(Alance AB, 2025)

Initially, during the first iteration of the prototype, the player would physically control each of the cameras using the mouse to look around and WASD to move the player. Reasons why the mouse controls the cameras rather than the direction of the player is because if you were to use the mouse to control the player's direction, it would make it difficult to navigate and to gain a sense of direction. Opting in to use keyboard controls to turn the player was a better decision as it made it easier and provides more control for the player.

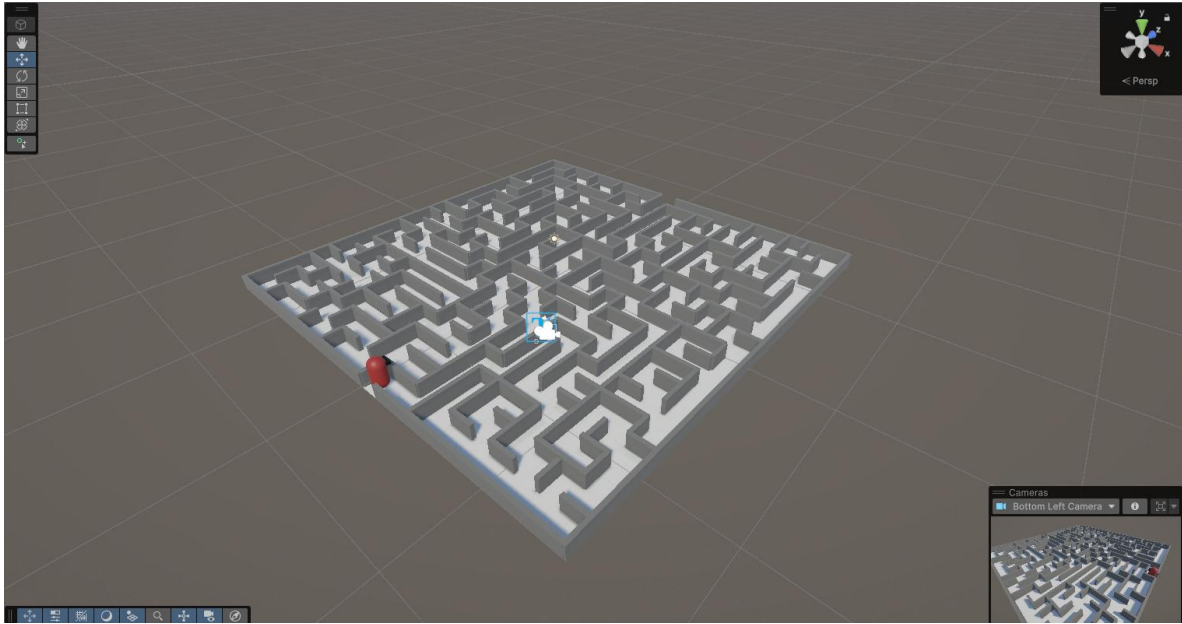


Figure 7 - Maze Idea Grey boxed

Although playing through the first iteration shown in **figure 7** looked promising, it was lacking the true aspect of 2PP which is to see yourself as the player through the eyes and perspective of another person or entity. Since the cameras have already been in place, the creative decision was to have a security officer look at the player through a computer that displays the cameras like a security guard looking at CCTV footage shown in **figure 8**.

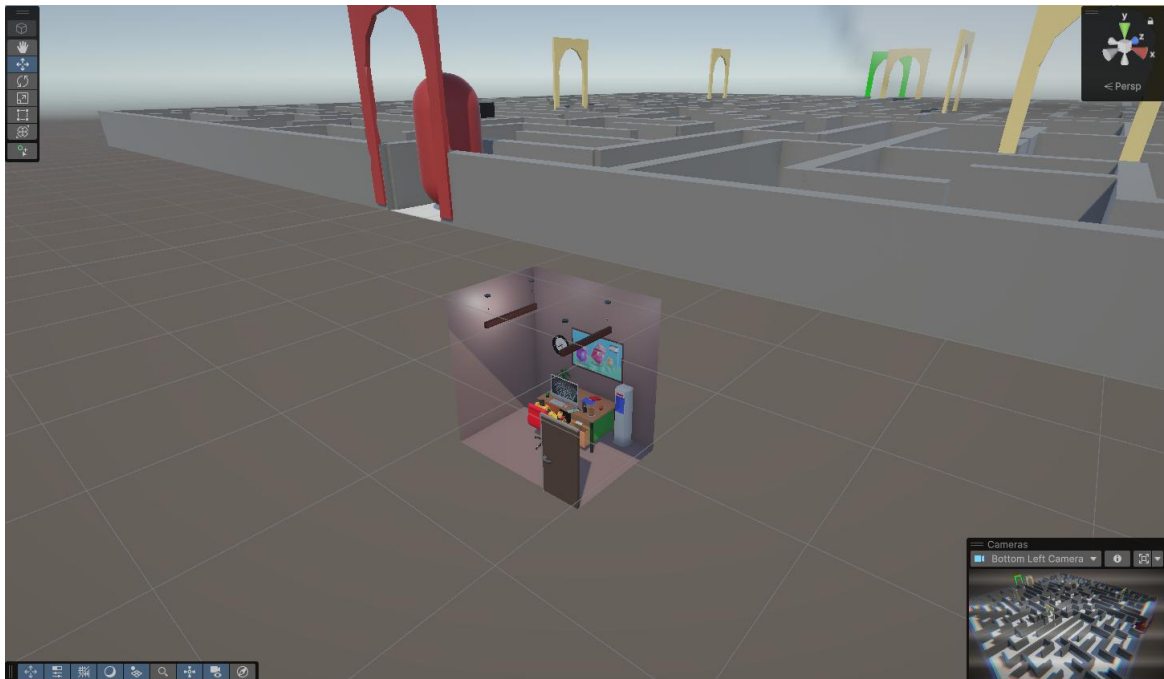


Figure 8 - Security Guard Office (Under Maze play area)

To emphasize the use of 2PP, a security guard office was created to create the vision that you are being observed from the perspective of the security guard by looking through a series of security cameras on the computer. For added creativity, post processing effects were used to create the illusion that the security guard was looking through a set of CCTVs Cameras observing the player. During gameplay, the player can use the mouse to look around the security office, using the scroll wheel to zoom in and out for better viewability of the screen and using numbers 1 – 4 to switch between cameras. To display each of the camera's output onto the security guard's computer, a render texture was used to project the view to make the feel of seeing yourself through CCTV a little bit more realistic, especially with post processing effects consisting of:

- Bloom
- Panini Projection
- Vignette
- Chromatic Aberration
- Film Grain



Figure 9 - Security Officer POV

During development and initial testing, it was quite difficult to navigate through the maze without some clear indication that you're going in the right direction. In the final prototype, gates were placed in some locations of the maze to help the player direct themselves towards the goal more easily. The red gate is the initial starting position, the yellow gates are for progression, and the green gate is the end goal.



Figure 10 - Overview of final Maze Prototype

To recap, the objective of the Maze prototype is to guide the red player to the end of maze by utilizing a series of different cameras to help you navigate through the maze. Each camera is displayed on a computer screen being viewed by a security guard. What makes this 2PP is that you are viewing yourself through the eyes of security guard who is observing you through the CCTV cameras. Using the **Observed Embodiment** term to briefly describe the point of view, you are **embodying** the security guard **observing** yourself on the computer screen.

3.2.3 Car Prototype

Next, the car prototype was in production. As the general idea for the rally driving instructor was plotted down on paper, the first thing to create was a drivable car for the player to control. Unity's documentation has a tutorial on creating a car with wheel colliders (Unity, 2025) and was used to create a drivable car for this prototype. Using the Synty prototype and City asset pack, the sports car was used for the drivable car as the other vehicle prefabs bundled with the city pack didn't work properly when using the wheel colliders, and the demo scene from the city pack was used for the player to drive around.



Figure 11 - Drivable Vehicle

The idea of being a rally driving instructor shifted whilst developing the prototype, mainly because of the vehicle model and the city scene used during development, influencing the decision. Within the critical context, Driver: San Francisco was one of the games that showcased 2PP within a driving genre by having the camera POV as the chaser driver rather than just behind the playable car, in **figure 12** shows the camera is positioned in the passenger seat observing the driver which the player controls.



Figure 12 - Passenger POV

Figure 13 showcases the viewpoint that the player sees when the game starts. The camera is situated in the passenger seat. The view controls act similarly to the maze prototype, using the mouse to look around the interior of the vehicle, right click to zoom in and scroll wheel to change the zoom FOV. However, to sell the fact that the player is the driver and not the passenger, a few keyframe animations were used to move the arms of the driver when the player tries to steer the vehicle. This method was primarily for showing that you are **embodying** the passenger, **observing** yourself who's the driver.

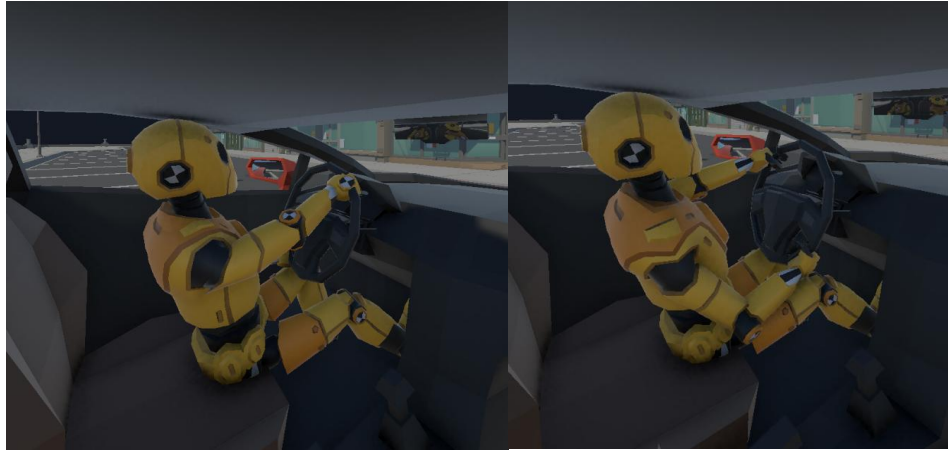


Figure 13 - Left and Right Steering

During testing, driving around in an empty city was boring to do, especially in a perspective that's uncommonly used. To combat this, a simple yet fun gameplay objective of chasing down another car and crashing into them was implemented. Since the car controller worked on the prototype asset car, the same vehicle model was used for testing and prototyping. The challenging factor was getting the vehicle to drive by itself independently, for simplicity, a waypoint system was developed for the AI driver to follow.



Figure 14 - AI Driver Vehicle & Properties

Looking at the waypoints, there are multiple points for the car to follow. The AI Driver looks ahead and finds the closest segment for it to follow. For this to work, it gets the list of waypoint's position and normalises the tangent to forward looking point, then as the car is driving when it reaches the minimum distance of the segment, the movement interpolates towards the next waypoint, creating a smoother driving experience.

At first, the car would drive at full speed at each waypoint, crashing into the buildings and other obstacles. To combat this, the car's speed was set to 30% its current max speed which worked for the most part. Although the car's speed was limited, there were times where the car would still steer off course and spin out, this was mainly due to the car taking each corner at full speed without braking. At each corner of the waypoint, the yellow debug spheres indicate breaking zones for the vehicle to reduce the throttle and take the corner slow just like a real driver. A final note, if the driver does get stuck on a building or an object within the world, it can reverse backwards to correct itself back onto the waypoint and continue following the path.

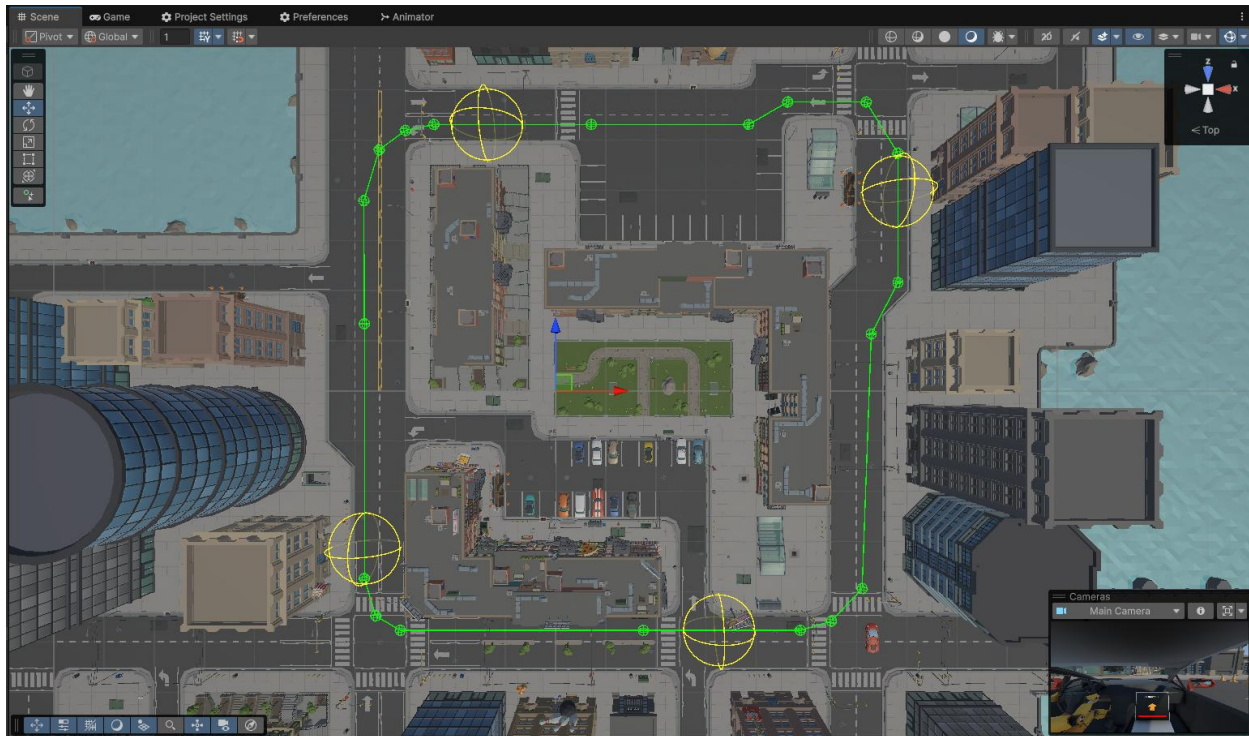


Figure 15 - Waypoint path

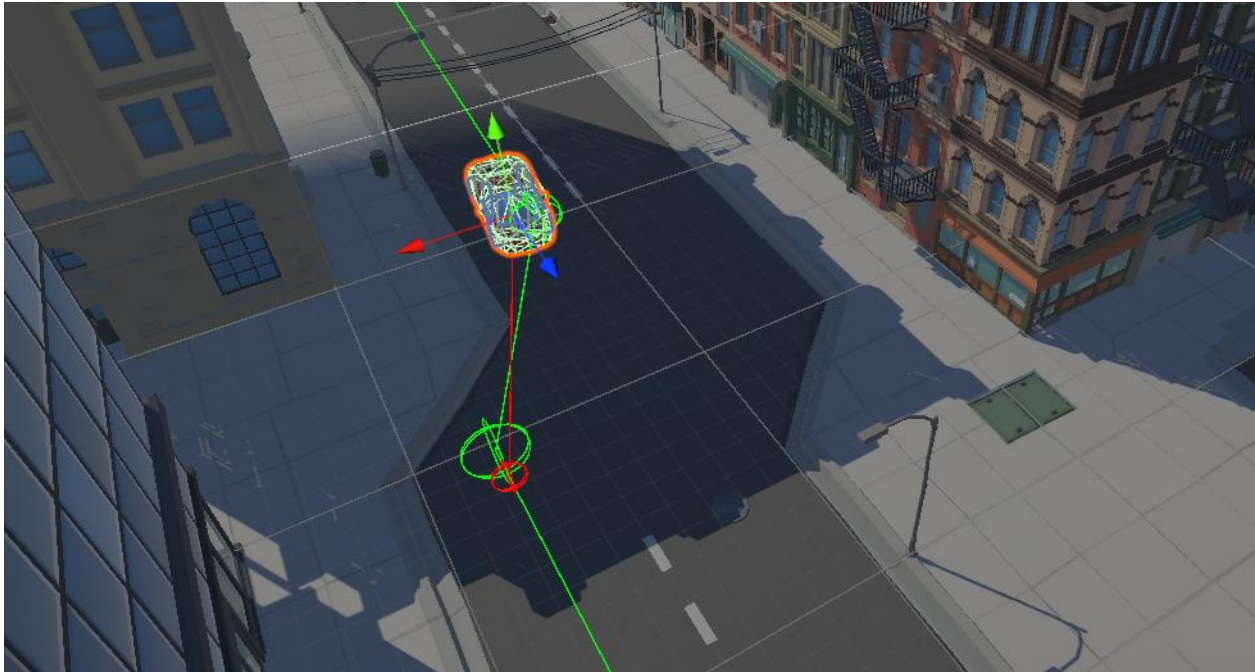


Figure 16 - Debug Gizmo showing car following waypoint

```

@ Unity Message | 0 references
void Update()
{
    if (waypoints == null || waypoints.Count == 0)
        return;

    Vector3 pos = transform.position;

    // --- Find the closest segment ---
    float minDist = float.MaxValue;
    int closestIndex = currentIndex;
    float closestT = 0f;
    for (int i = 0; i < waypoints.Count; i++)
    {
        Vector3 a = waypoints[i].position;
        Vector3 b = waypoints[(i + 1) % waypoints.Count].position;
        Vector3 ab = b - a;
        float t = Mathf.Clamp01(Vector3.Dot(pos - a, ab.normalized) / ab.magnitude);
        Vector3 closestPoint = Vector3.Lerp(a, b, t);
        float dist = (pos - closestPoint).sqrMagnitude;
        if (dist < minDist)
        {
            minDist = dist;
            closestIndex = i;
            closestT = t;
        }
    }
    currentIndex = closestIndex;

    Transform wpA = waypoints[currentIndex];
    Transform wpB = waypoints[(currentIndex + 1) % waypoints.Count];

    // --- Dynamic lookahead distance based on speed ---
    float speed = rb.linearVelocity.magnitude;
    float dynamicLookahead = Mathf.Lerp(lookaheadDistance, lookaheadDistance + 3f, Mathf.InverseLerp(0, maxSafeSpeed, speed));

    // --- Use improved lookahead ---
    Vector3 lookaheadPoint = GetLookaheadPoint(pos, currentIndex, closestT, dynamicLookahead);

    // Steering calculation
    Vector3 localTarget = transform.InverseTransformPoint(lookaheadPoint);
    float angleToTarget = Mathf.Atan2(localTarget.x, localTarget.z) * Mathf.Rad2Deg;
    float normalizedSteering = Mathf.Clamp(angleToTarget / maxSteeringAngle, -1f, 1f);

    // --- Path direction (for spin-out detection) ---
    Vector3 pathDir = (lookaheadPoint - pos).normalized;
    Vector3 carForward = transform.forward;
    float alignment = Vector3.Dot(carForward, pathDir); // 1 = aligned, -1 = opposite
}

```

Code Snippet 1- WaypointFollower.cs Lookahead formula

```

WaypointFollower.cs
Assembly-CSharp
WaypointFollower
GetLookaheadPoint(Vector3 pos, int startIndex, float tOnSegment, float lookaheadDist)

169         break;
170     }
171
172     if (nearBrakePoint)
173     {
174         // --- Throttle logic for braking based on speed in mph ---
175         float speedMph = aiDriver.CurrentSpeedMph;
176         if (speedMph > brakePointTargetSpeedMph)
177         {
178             aiDriver.throttle = -1f; // full brake/reverse
179         } else if (speedMph > brakePointTargetSpeedMph * 0.5f)
180         {
181             aiDriver.throttle = -0.5f; // moderate brake
182         } else
183         {
184             aiDriver.throttle = 0.1f; // coasting/stop
185         }
186         aiDriver.steering = Mathf.Lerp(aiDriver.steering, normalizedSteering, Time.deltaTime * steeringSmoothing);
187         return;
188     }
189
190     // --- Normal driving logic ---
191     aiDriver.steering = Mathf.Lerp(aiDriver.steering, normalizedSteering, Time.deltaTime * steeringSmoothing);
192
193     // --- Predict Upcoming Turn ---
194     int nextIndex = (currentIndex + 1) % waypoints.Count;
195     int nextNextIndex = (currentIndex + 2) % waypoints.Count;
196     Vector3 nextA = waypoints[nextIndex].position;
197     Vector3 nextB = waypoints[nextNextIndex].position;
198     Vector3 dirCurrent = (wpB.position - wpA.position).normalized;
199     Vector3 dirNext = (nextB - nextA).normalized;
200     float bendAngle = Vector3.Angle(dirCurrent, dirNext);
201
202     // --- Speed & Braking Control ---
203     float speedFactor = Mathf.InverseLerp(minSafeSpeed, maxSafeSpeed, speed);
204     float turnAngleFactor = Mathf.InverseLerp(turnsLowdownAngle, fullBrakeAngle, Mathf.Max(Mathf.Abs(angleToTarget), bendAngle));
205     float totalBrakeFactor = Mathf.Clamp01(Mathf.Lerp(turnAngleFactor, speedFactor, speedWeight));
206     float targetThrottle = Mathf.Lerp(baseThrottle, minThrottle, totalBrakeFactor);
207
208     // Smooth transitions
209     aiDriver.throttle = Mathf.Lerp(aiDriver.throttle, targetThrottle, Time.deltaTime * 3f);
210
211 }
212
213 // references
214 Vector3 GetLookaheadPoint(Vector3 pos, int startIndex, float tOnSegment, float lookaheadDist)
215 {
216     int segA = startIndex; // current segment
217     int segB = (segA + 1) % waypoints.Count; // next segment
218     Vector3 a = waypoints[segA].position; // start of current segment
219     Vector3 b = waypoints[segB].position; // end of current segment
220     Vector3 ab = b - a; // vector of current segment
221
222     float segLength = ab.magnitude; // length of current segment
223     float distOnSeg = (1f - tOnSegment) * segLength; // remaining distance on current segment
224
225     if (lookaheadDist <= distOnSeg) // lookahead point is on current segment
226     {
227         float lookaheadT = tOnSegment + (lookaheadDist / segLength); // normalized t on current segment
228         return Vector3.Lerp(a, b, lookaheadT); // interpolate to get lookahead point
229     } else
230     {
231         float remaining = lookaheadDist - distOnSeg; // remaining distance to lookahead point
232         return GetLookaheadPoint(pos, segB, 0f, remaining); // recurse to next segment
233     }
234 }

```

Code Snippet 2 - WaypointFollower.cs Driving Logic

Getting into the primary objective of chasing the AI driver and crashing into them, because of the low visibility inside the interior, a laptop was created with an indicator arrow pointing towards the AI driver, so the player knows where they are. A health bar was also added to indicate how much health the car has. Initially at first, the health would be indicated visually with smoke gradually increasing from the vehicle's engine shown in **figure 18** but it was difficult to know how much health the car had, so the bar was added for better clarity.



Figure 17 - Primary Objective POV

Speaking of visual health, the car has 3 stages of indicating vehicle health. At 75% health, the car starts showing a small puff of smoke coming from the rear of the vehicle (model based on a Lamborghini, the engine is positioned at the rear), at 50% health the smoke starts growing more, and finally at 25% health there's a lot of smoke coming out of the vehicle. Lastly, when the vehicle's health diminishes, the vehicle model swaps for a blacked-out version with fire particles and an explosion effect playing to indicate the vehicle has exploded and no longer functions.



Figure 18 - 75% | 50% | 25% Health



Figure 19 - Vehicle at 0% health

3.2.4 Boss Prototype

Lastly, the final prototype based on the boss rush sketches was developed. Within the sketches, the main idea was to fight a multitude of different bosses with different effects and weaknesses that the player would exploit to defeat them, but the main factor was to focus on seeing the player through the perspective of the enemy boss. The primary gameplay for this prototype is the enemy boss is firing their weapon towards the player, and the player must fire back at the boss to eliminate them. This scenario takes direct inspiration from the Den Mother boss fight featured in Psychonauts as explained in the critical context.

Figure 20 shows 3 predominant platforms for the main antagonist and a central arena for the player to run around. Within the central arena are multiple obstacles for the player to run around but also to hide behind each obstacle to avoid direct fire and break the line of sight from the enemy. The player and the enemy are both equipped with a weapon that shoots bullet projectiles and the boss jumps between 3 different points on the level, the middle, right and left and alternates between each platform.

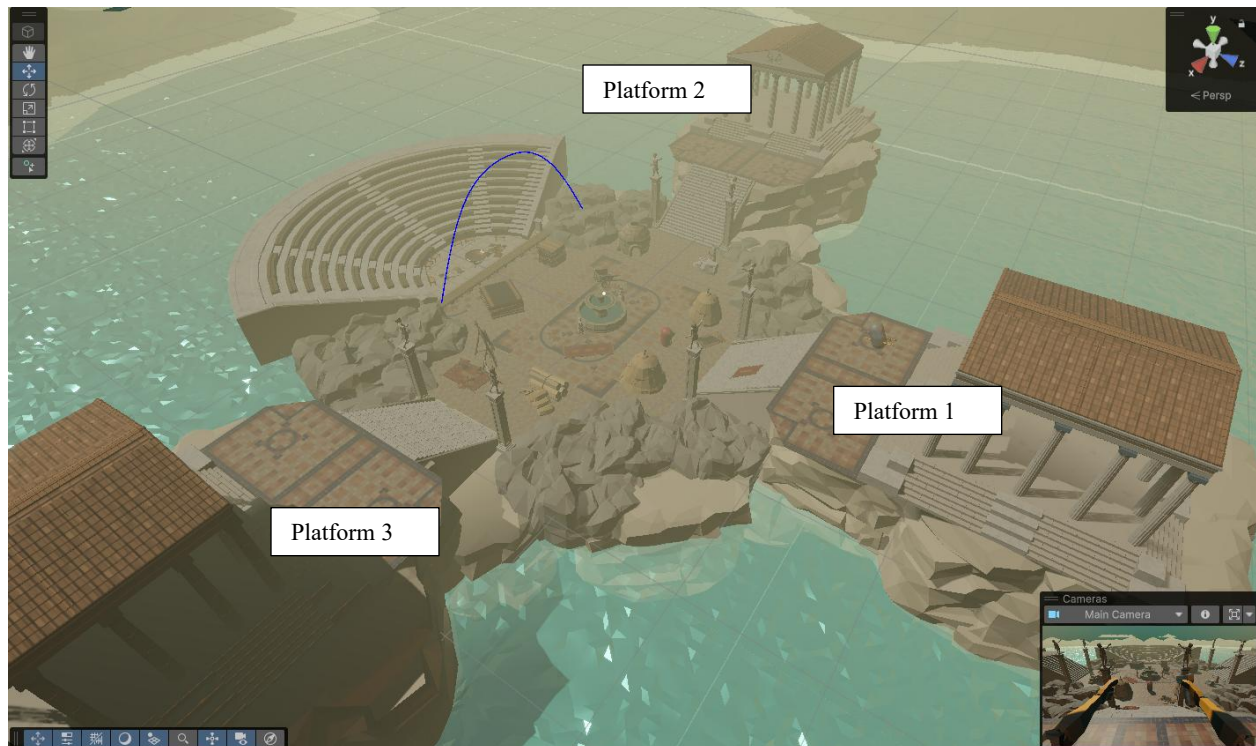


Figure 20 - Boss Arena

Focusing on the player's movement, the movement is the same as the Maze prototype however the one change is the how the player shoots the weapon. During normal movement, the player cannot fire their weapon until they enter battle mode. When holding down the right mouse button, the player enters battle mode which points the direction of the player towards the enemy and by clicking or holding down left click, the player can fire the weapon projectiles to damage the enemy.

To showcase the use of 2PP, the camera is positioned in the eyes of the antagonist, and the camera follows the position of the player, so they are always within frame of the camera. The POV of the boss shows their arms and a weapon in the right hand to emphasise that although you control the red capsuled player, you are embodying the boss and seeing the world through their eyes. You are **embodying** the boss, **observing** yourself on the arena.



Figure 21 - Game & Boss POV

During development, the gameplay seemed a little too bland of hiding behind obstacles and peeking out to damage the boss, so a couple of implementations were made. Both the boss and the player have their own health bar, to encourage the player to hide behind obstacles, a regenerative health mechanic was implemented so if the player took enough damage they can hide and regain their health over a small amount



Figure 22 - Boss POV at Platform 2 & 3

of time. By encouraging this type of behaviour for the player, a gameplay mechanic was put in place for the boss. The boss shoots a grid of 10 debug rays from the camera's height to detect if the player is within view of the boss. Out of the 10 rays, if the player is within view, the rays light up green and marks that the player is visible to the enemy, else the ray is coloured yellow. If the player moves behind cover and the enemy cannot see the player, then the rays will be coloured red to indicate that the player is no longer visible. With this, if the player is not visible for 5 seconds, the boss jumps to a different platform within the level to flank the player and to be able to attack, providing challenge for the player. All shown in **figures 22, 23 & 24**.


```

TargetPlayer.cs
Assembly-CSharp
TargetPlayer

107 for (int y = 0; y < grid; y++)
108 {
109     float v = (grid == 1) ? 0.5f : (float)y / (grid - 1);
110     float pitch = Mathf.Lerp(-coneAngle / 2f, coneAngle / 2f, v);
111
112     for (int x = 0; x < grid; x++)
113     {
114         if (raysDrawn >= rayCount)
115             break;
116
117         float u = (grid == 1) ? 0.5f : (float)x / (grid - 1);
118         float yaw = Mathf.Lerp(-coneAngle / 2f, coneAngle / 2f, u);
119
120         Quaternion rot = Quaternion.Euler(pitch, yaw, 0);
121         Vector3 dir = rot * forward;
122
123         Vector3 currentOrigin = origin;
124         float remainingLength = rayLength;
125         bool hitSomething = false;
126         bool hitPlayerThisRay = false;
127         float hitDistance = rayLength;
128
129         while (remainingLength > 0f)
130         {
131             Ray ray = new Ray(currentOrigin, dir);
132             RaycastHit hit;
133
134             if (Physics.Raycast(ray, out hit, remainingLength))
135             {
136                 hitSomething = true;
137                 bool ignore = false;
138                 foreach (var tag in ignoredTags)
139                 {
140                     if (hit.collider.CompareTag(tag))
141                     {
142                         currentOrigin = hit.point + dir * 0.01f;
143                         remainingLength -= hit.distance + 0.01f;
144                         ignore = true;
145                         break;
146                     }
147                 }
148                 if (ignore)
149                 {
150                     continue;
151                 }
152                 if (hit.collider.CompareTag("Player"))
153                 {
154                     hitPlayerThisRay = true;
155                     hitDistance = rayLength - remainingLength + hit.distance;
156                     playerInView = true;
157                 }
158                 else
159                 {
160                     hitDistance = rayLength - remainingLength + hit.distance;
161                 }
162                 break;
163             }
164             else
165             {
166                 break;
167             }
168         }
169
170         if (hitPlayer != null && raysDrawn < hitPlayer.Length)
171         {
172             hitPlayer[raysDrawn] = hitPlayerThisRay;
173             rayDirections[raysDrawn] = dir;
174             rayHitDistances[raysDrawn] = hitDistance;
175             rayHitSomething[raysDrawn] = hitSomething;
176         }
177     }
178     raysDrawn++;
179 }

```

Code Snippet 3 - TargetPlayer.cs



Figure 23 - Debug Rays detecting Player is in view



Figure 24 - Debug Rays detecting Player not in view

Lastly, to add more challenge to the prototype, the boss has 3 different states depending on how much health is left. Like the car prototype with the AI Driver's health, the bullet speed increases and reload speed decreases relative to how much health is remaining, so if the player has dealt 25% of the boss' health, this triggers state 1 and the bullet speed and reload speed are affected by 25%. State 2 happens when 50% of the health has been dealt and finally state 3 is 75%.

```

BossHealth.cs
Assembly-CSharp
BossHealth

41
42
43 1 reference
44 public void SetHealth(float value)
45 {
46     health = Mathf.Clamp(value, 0f, maxHealth);
47     if (healthBar != null)
48         healthBar.fillAmount = health / maxHealth;
49 }
50
51 @ Unity Message | 0 references
52 void Update()
53 {
54     health = Mathf.Clamp(health, 0f, maxHealth);
55
56     if (health != lastHealth && healthBar != null)
57     {
58         healthBar.fillAmount = health / maxHealth;
59     }
60
61     if (bulletProjectile != null)
62     {
63         if (health < 0.25f * maxHealth && enrageState != 3)
64         {
65             bulletProjectile.bulletSpeed = originalBulletSpeed * 1.75f;
66             bulletProjectile.reloadTime = originalReloadTime * 0.25f;
67             enrageState = 3;
68         }
69         else if (health < 0.5f * maxHealth && enrageState != 2 && health >= 0.25f * maxHealth)
70         {
71             bulletProjectile.bulletSpeed = originalBulletSpeed * 1.5f;
72             bulletProjectile.reloadTime = originalReloadTime * 0.5f;
73             enrageState = 2;
74         }
75         else if (health < 0.75f * maxHealth && enrageState != 1 && health >= 0.5f * maxHealth)
76         {
77             bulletProjectile.bulletSpeed = originalBulletSpeed * 1.25f;
78             bulletProjectile.reloadTime = originalReloadTime * 0.75f;
79             enrageState = 1;
80         }
81         else if (health >= 0.75f * maxHealth && enrageState != 0)
82         {
83             bulletProjectile.bulletSpeed = originalBulletSpeed;
84             bulletProjectile.reloadTime = originalReloadTime;
85             enrageState = 0;
86         }
87     }
88
89     if (health <= 0)
90     {
91         Debug.Log("Boss has died");
92         if (pauseMenu != null)
93         {
94             pauseMenu.ShowVictoryScreen();
95         }
96     }
97
98     lastHealth = health;
99 }

```

Code Snippet 4 - BossHealth.cs

3.2 Phase 3 - Prototype Evaluation

During August 5th to August 8th, a playtest was conducted to gather useful feedback and information based on the maze, car and boss prototypes that have been created. This section will detail how the playtest was structured. All playtest documents are available in [Appendix 2](#).

3.2.1 Recruitment

As the three base prototypes were created, participants were recruited online using a screener survey to assist in playing the prototypes. **Figure 25** shows the poster with a QR sign up code was also created to be posted up around the campus in case anyone was interested in participating. The screener survey asked general questions such as:

- Name
- Age Group
- How often you play games
- Type of games played
- Preferred Perspective in games
- Willing to travel to campus
- How to be contacted
 - Email
 - Phone
- Suitable Time
- Invite to future testing
- Consent for participation

It was challenging trying to recruit participants to help and play test the prototypes that were made, although a total of 4 participants were able to attend the play test sessions to provide their thoughts and feedback. Each participant had their own time slot during the first week of August which was best suited for them.



Figure 25 - Playtest Poster

3.2.2 Environment and Equipment

A week was booked inside the Interaction Lab at City, University to run the prototype on the lab computers, using the existing software that was installed onto the system like OBS Studio to capture and record the audio, webcam footage and the screen. A runnable executable was compiled and installed onto the lab PC to ensure everything ran smoothly. Playable prototype is available in [Appendix 3](#).

Using OBS, each participant was recorded so that the footage could be reviewed afterwards in case any comments and thoughts were said during the playtest that were missed when writing up notes as they were testing the build. Each of the recordings were transferred to OneDrive and would be kept securely for up to 10 years as per the University policy.

Interaction Lab PC Specs:

- Intel Core i7-14700K
- 32GB RAM
- GPU: Nvidia GeForce RTX 4060 8GB

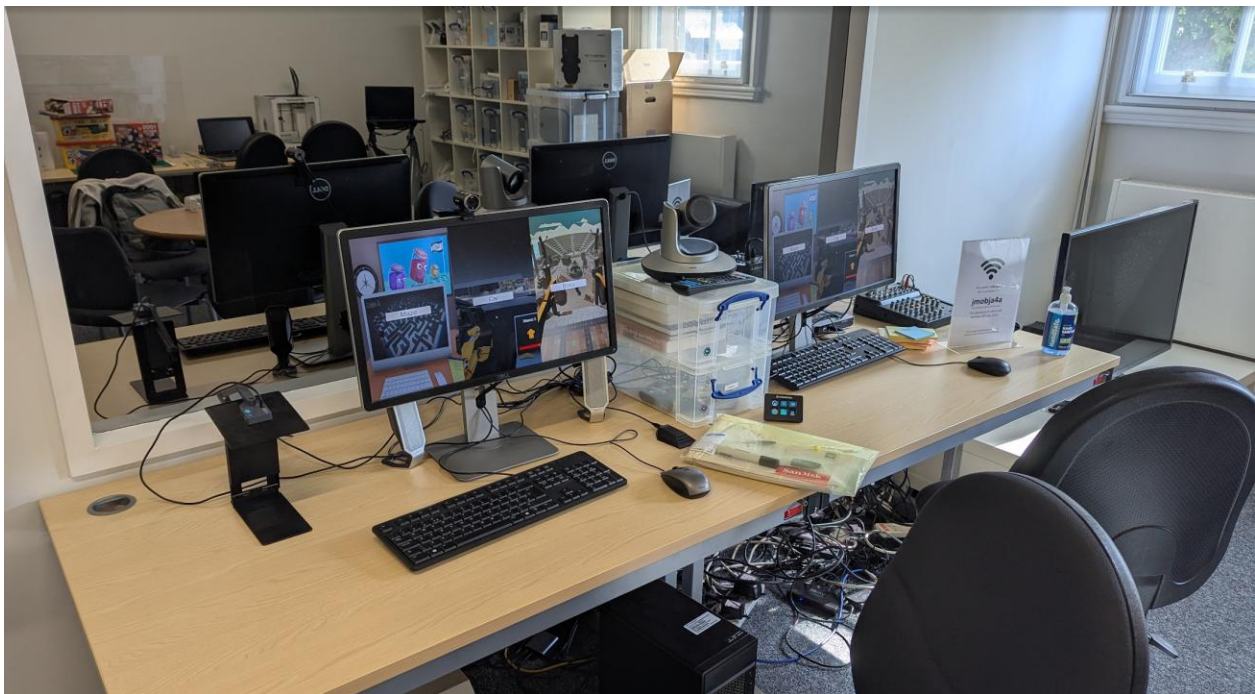


Figure 26 - Interaction Lab Playtest Environment

3.2.3 Consent

All the participants session were provided with a consent form to fill out. Before signing, a playtest information sheet was provided to the participant for them to read and any questions they had about the testing were answered.

3.2.4 Playtest Session Structure

Each session lasted approximately 45 minutes, and I used a Google form to keep track of answers provided during the interview phase. Below is the structure of the playtest session:

- Welcome and Introduction – Greeting participants and obtaining consent (Approx. 5 minutes)
- Pre Interview Questions – Discussing what games they like as well as their general understand of the second person perspective and if they knew what it meant. (1 minute)
- Playing the prototype – Participants would first play the scenario, and then afterwards complete the IEQ survey tied to the scenario. This was iterated for each subsequent scenario. (Approx. 30 minutes)
- Post Interview Questions – Gathering insight about their views and understanding of 2PP, which prototype scenario best showcased 2PP, and if 2PP is more of a gimmick or has long term potential (Approx. 10 minutes).

Pre Interview Questions

At the start of the playtest, the pre interview was to gather some insight as to what kind of person the participant to see if they have had any understanding relating to 2PP within games.

- Prior to the study, have you encountered any games that have used 2PP?
- Before we start the user testing, do you know what ‘2PP’ means in games?

Prototype Scenario Tasks

Within each prototype, the player has a one main objective to complete within the demo. This is to observe how they tackle the objective from a different perspective.

Table 2 - Scenario Objectives

Scenario	Objective
Maze	You are stuck in a Maze and must navigate through it to complete the maze.
Car	You are driving and the objective is to crash into the NPC Driver and destroy their car
Boss	You must defeat the enemy that is shooting at you. As the enemy loses more health, their attack becomes stronger and faster. Use your surroundings to take cover and recover health.

Prototype IEQ Questions

When the participant has completed each scenario, they filled out a survey that was created using questions from the IEQ. There was a total of 23 questions, however a couple of extra questions were added that best fit the theme of 2PP: (Drannalcox, 2024)

- To what extent did you feel like ‘Yourself’ in the game environment?
- Did the Camera / Viewpoint ever make it harder to play or understand the objective?

Post Interview Questions

After the participant has completed each prototype scenario and has filled out the IEQ survey, the post interview is to gather feedback based on their experience interacting with a couple of game genres designed for 2PP.

- Were you able to identify how each level showcased 2PP?
- After playing all scenarios, how would you describe 2PP in games?
- From what you played, what differentiates 2PP from first & third person?
- Which prototype scenario best demonstrated 2PP?
- How did that scenario demonstrate 2PP better than the other choices?
- Did any of the prototypes feel like something you’ve played before?
- Have your views and understanding of 2PP changed after testing out each prototype?
- Could you imagine 2PP being used effectively in other game genres?
- Would you be interested in playing a full game using only 2PP?
- Do you see 2PP as a novel gimmick or something that has long term potential?

3.2.5 Use of 2PP in Games survey

After completing the first playtest session, not enough data was gathered when asking what type of game people would like to see in the second person after testing the Phase 2 prototypes. A Google form was created and distributed to online forums to gather as much data as possible to generate ideas. Using some of the games discussed in the critical context, the structure of the form was as followed:

- Video Clip – Participant would watch a short clip of a game that uses 2PP
 - Driver: San Francisco
 - Psychonauts
 - Second Person Shooter Zato
 - Call of Duty: MW19
- Whose perspective was shown in the clip?
- Write in detail what you **liked** | **Disliked** | **Unique** about in the video clip?
- If you could think about game ideas, you'd like to see relating to this video clip created in the second person perspective, please write them down. Be as descriptive as you can.

A total of 12 responses were gathered. One response stood out that would inspire that next prototype to be built to implement a second person perspective mentions “The game I really enjoyed, was the Road 96, which is one of the best second person perspective IMO”. Looking at screenshots and gameplay of Road 96, “Road 96 is an interactive story that uses the first-person perspective and random generation to mix and match the sequence of events you will experience. You play as a random set of teenagers with the goal of surviving each sequence in order to reach the border and figuring out the best way to cross by judging the condition and resources of your current character.”(*Fandom*, n.d.) An image of a reporter standing in front of police cars stood out and provided the idea of using the camera man's perspective as the main game camera while following around a journalist. This idea would become the next prototype for this project.

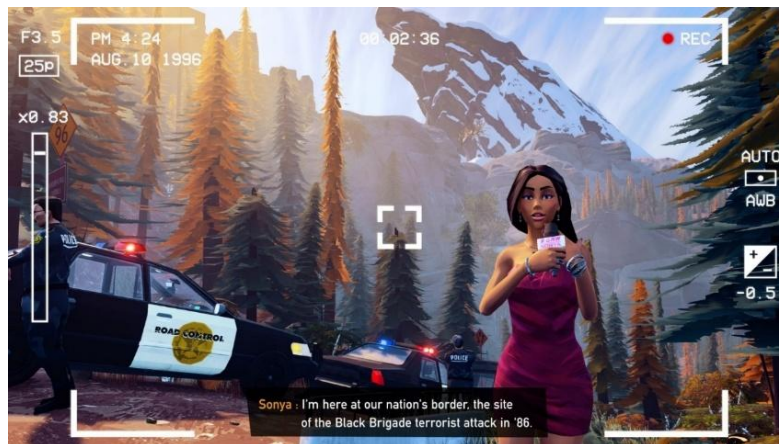


Figure 27 - Road 96

3.4 Phase 4 – Final Prototype: Journalist

The final prototype to be built was the Journalist prototype. From the techniques and implementations learned from developing the prior demo's, one thing that was key to really emphasise the perspective of second person is to have the camera embody someone else and to have the main player within centre frame of the scene consistently. As the game features a camera man and the reporter, the design choice was to have the camera man hold a smartphone and to show the hands to further display that although you are controlling the reporter, you are **embodying** the camera man, **observing** and controlling the news reporter.



Figure 28 - Camera Man's POV

Context for this prototype is you're a news reporter and heard news about a crime that has been committed in the City's Park. Your job is to discover and find clues around the park, reporting your findings with your camera man whilst sneaking past the roaming guards trying not to get caught. Each clue has a film camera object to identify that it can be reported and requires a certain amount of main footage to be recorded. A feature called "B Roll" which is often seen in news casts is footage capturing the main subject of the clue, not filming the news reporter. Both require capturing footage before marking the report complete.



Figure 29 - Overview of Park

When the player enters the trigger zone and presses right click, they enter “Reporter Mode”. When the player is in this mode, their movement is disabled, and the player’s animation moves the microphone to their mouth to indicate they are reporting the clue they’ve found. Whilst reporting, a high-pitched voice starts playing meaning that the player is speaking and must record however long the prompt tells them to. If the report requires B Roll footage, the player presses the B button to move the camera man to focus the camera on the subject in focus, although for the B Roll footage to count, the player must use the scroll wheel to zoom in the camera as there is a zoom threshold to complete.

While the player is reporting, if a guard sees the player, the movement is enabled and a prompt “You’re currently being chased” is displayed, meaning the player cannot report at any of the areas until they aren’t being chased anymore. All is shown in **figures 30, 31 & 32**.



Figure 30 - Entering Report Trigger Box



Figure 31 - Reporter Mode

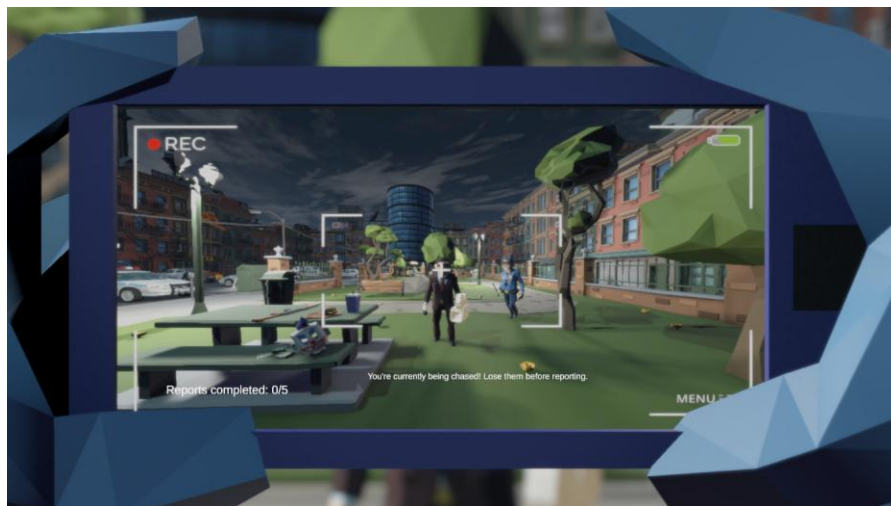


Figure 32 - Seen by Guard

Whilst creating the basic gameplay flow of this prototype, the thought of adding obstacles such as guards were added to create the feeling of the player infiltrating a closed crime scene after hours and must avoid getting caught to record for their news reel. Using Unity's Navmesh system, these enemy guards can move on their own independently and avoid obstacles accordingly based on the baked Navmesh floor. The enemy uses a state machine to transition between a different set of states, depending on if the player is in view of the guard.

The **patrol state** has the guard follow a set of waypoints (like the AI Driver system) to follow. The Navmesh Agent moves along the path from one waypoint to another and stops at each point for a few seconds before walking along to the next point. Once the guard has reached the final point in the list, it goes back to the first point completing the loop. In Unity, the waypoints are displayed as diamonds with a white line to display the circuit that the Navmesh agent can follow.



Figure 33 - Guard Navmesh Pathing


```

1  using UnityEngine;
2
3  7 references
4  public class PatrolState : BaseState
5  {
6      public int waypointIndex; // track which waypoint we are currently targetting
7      public float waitTimer; // how long we have waited at a waypoint
8
9      2 references
10     public override void Enter()
11     {
12         // Ensure we have a valid path and waypoints
13         if (enemy.path == null || enemy.path.waypoints == null || enemy.path.waypoints.Count == 0)
14         {
15             Debug.LogWarning($"[{nameof(PatrolState)}] Enemy '{enemy.name}' has no path/waypoints assigned.");
16             return;
17         }
18
19         // Make sure the agent can move
20         if (enemy.Agent != null)
21         {
22             enemy.Agent.isStopped = false;
23         }
24
25         // Clamp index and pick a sensible starting waypoint (nearest), then set destination
26         waypointIndex = Mathf.Clamp(waypointIndex, 0, enemy.path.waypoints.Count - 1);
27
28         int nearest = 0;
29         float minSqr = float.MaxValue;
30         for (int i = 0; i < enemy.path.waypoints.Count; i++)
31         {
32             var wp = enemy.path.waypoints[i];
33             if (wp == null) continue;
34             float sqr = (enemy.transform.position - wp.position).sqrMagnitude;
35             if (sqr < minSqr)
36             {
37                 minSqr = sqr;
38                 nearest = i;
39             }
40         }
41
42         waypointIndex = nearest;
43
44         // If we're basically on top of the nearest, move to the next one to keep moving
45         if (Vector3.Distance(enemy.transform.position, enemy.path.waypoints[waypointIndex].position) < 0.5f)
46         {
47             waypointIndex = (waypointIndex + 1) % enemy.path.waypoints.Count;
48         }
49
50         enemy.Agent.SetDestination(enemy.path.waypoints[waypointIndex].position);
51         waitTimer = 0f;
52     }
53 }

```

Code Snippet 5 - PatrolState.cs

If the player enters the guard's field of view, the guard switches to the **chase state**. Within this state, the enemy guard will chase the player around. This is done by using a ray to detect if the player is within the eye range and has a set distance to how far the guard can see. There are two methods for the guard to exit this state and return to the patrol state. One is if the player is far enough away from the guard, the AI waits a little before switching, and the second method is if the player runs out of the guard's patrol area (example the park), the state switches. Whilst the guard is chasing the player, to add some gameplay risk, if the guard reaches the player within a small certain radius, the guard can capture the player, forcing them to respawn outside the area. If the player does get captured and respawns, the states get reset back to patrol.

```

ChaseState.cs  x PatrolState.cs  StateMachine.cs
Assembly-CSharp  ChaseState

29      enemy.StopChaseAudio();
30      enemy.RestoreOriginalStoppingDistance();
31
32      // End chase music crossfade (reference counted across multiple chases / enemies)
33      BackgroundMusic.Instance?.EndChase();
34
35      // ADD: notify global listeners that a chase ended
36      Enemy.SignalChaseEnd();
37  }
38
39  2 references
40  public override void Perform()
41  {
42      if (enemy.IsProcessingCatch) return;
43
44      // Distance-based catch
45      if (enemy.Player != null)
46      {
47          Vector3 diff = enemy.Player.transform.position - enemy.transform.position;
48          if (diff.sqrMagnitude <= sqrCatchDistance)
49          {
50              enemy.CatchPlayer();
51              return;
52          }
53      }
54
55      // Abort if player leaves guard area
56      if (!enemy.IsPlayerInGuardArea)
57      {
58          stateMachine.ChangeState(new PatrolState());
59          return;
60      }
61
62      // Maintain pursuit / memory
63      if (enemy.CanSeePlayer())
64      {
65          lostSightTimer = 0f;
66          if (enemy.Agent != null && enemy.Player != null)
67          {
68              enemy.Agent.SetDestination(enemy.Player.transform.position);
69              enemy.LastKnownPos = enemy.Player.transform.position;
70          }
71      }
72      else
73      {
74          if (enemy.Agent != null)
75              enemy.Agent.SetDestination(enemy.LastKnownPos);
76
77          lostSightTimer += Time.deltaTime;
78          if (lostSightTimer >= lostSightTimeout)
79          {
80              stateMachine.ChangeState(new PatrolState());
81          }
82      }
83  }

```

Code Snippet 6 - ChaseState.cs

```

Unity Message | 0 references
private void Update()
{
    CanSeePlayer();
    if (stateMachine != null && stateMachine.activeState != null)
        currentState = stateMachine.activeState.ToString();
}

5 references
public bool CanSeePlayer()
{
    if (player == null) return false;
    if (Vector3.Distance(transform.position, player.transform.position) >= sightDistance) return false;

    Vector3 targetDirection = player.transform.position - transform.position - (Vector3.up * eyeHeight);
    float angleToPlayer = Vector3.Angle(targetDirection, transform.forward);
    if (angleToPlayer < -fieldOfView || angleToPlayer > fieldOfView) return false;

    Ray ray = new Ray(transform.position + (Vector3.up * eyeHeight), targetDirection);
    if (Physics.Raycast(ray, out RaycastHit hitInfo, sightDistance))
    {
        if (hitInfo.transform.gameObject == player)
        {
            Debug.DrawRay(ray.origin, ray.direction * sightDistance, Color.red);
            return true;
        }
    }
    return false;
}

```

Code Snippet 7 - ChaseState.cs CanSeePlayer() Method



Figure 34 - Guard Ray chasing and following Player

Lastly, the **search state** is triggered when the player activates a distraction or by whistling. If the player has activated the distraction, then the guard moves towards the position of the distraction. During the search state, the search duration could last between 5 and 12 seconds and searches within a set radius. These timings were used to add some randomness and variety to the guard's search timer before switching back to the **patrol state** if the player is nowhere to be found.

```

17 // head to last known player position
18 enemy.Agent.SetDestination(enemy.LastKnownPos);
19
20 // ADD: initialize timers and one-time thresholds
21 searchTimer = 0f;
22 moveTimer = 0f;
23 searchDuration = Random.Range(5f, 12f);
24 waitBetweenMoves = Random.Range(3f, 5f);
25 }
26
27 public override void Perform()
28 {
29     // Do not search outside the guard area
30     if (!enemy.IsPlayerInGuardArea)
31     {
32         stateMachine.ChangeState(new PatrolState());
33         return;
34     }
35
36     // If we see the player, immediately chase
37     if (enemy.CanSeePlayer())
38     {
39         stateMachine.ChangeState(new ChaseState());
40         return;
41     }
42
43     // Robust "arrived" check (match PatrolState tolerance)
44     if (!enemy.Agent.pathPending &&
45         enemy.Agent.remainingDistance <= enemy.Agent.stoppingDistance + 0.1f)
46     {
47         searchTimer += Time.deltaTime;
48         moveTimer += Time.deltaTime;
49
50         // After waiting a bit, move to a nearby random point
51         if (moveTimer > waitBetweenMoves)
52         {
53             // Keep movement roughly on the horizontal plane
54             Vector3 randomOffset = Random.insideUnitSphere * 10f;
55             randomOffset.y = 0f;
56
57             enemy.Agent.isStopped = false;
58             enemy.Agent.SetDestination(enemy.transform.position + randomOffset);
59
60             moveTimer = 0f;
61             waitBetweenMoves = Random.Range(3f, 5f); // ADD: resample per hop
62         }
63
64         // Return to patrol after the total search duration
65         if (searchTimer > searchDuration)
66         {
67             stateMachine.ChangeState(new PatrolState());
68         }
69     }
70 }

```

Code Snippet 8 - SearchState.cs

Enemy State Machine

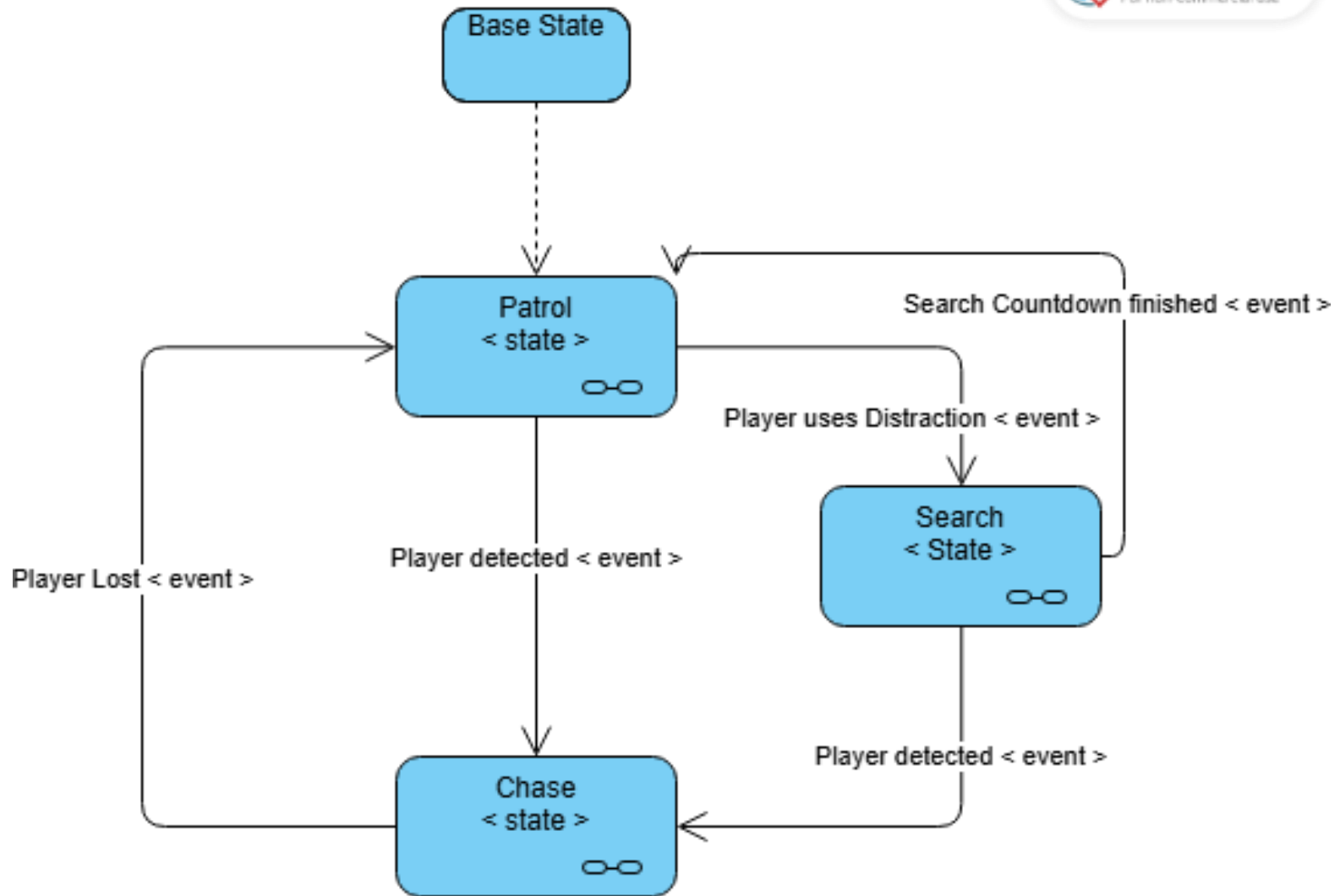


Figure 35 - Enemy State Machine

4 Results

This section of the report discusses the results collected from running playtest sessions, affinity diagrams representing the data as to what each participant did during playtesting, the IEQ scores and factors including: Challenge, Control, Real World, Emotional and Cognitive Involvement and lastly what was achieved with the final prototypes.

4.1 Playtest Session 1

As mentioned in the method, the first playtest session was hosted during August 5th – 8th with the help of 4 participants testing out the 3 prototype scenarios created at the time which were Maze, Car and Boss. Each participant filled out the IEQ survey corresponding which scenario they played, and each score affected the immersion value that's assessed across five dimensions:

- **Challenge:** How well the difficulty of the task matches the individual's skills.
- **Control:** How much autonomy and ease the individual feels during the activity.
- **Real-World Dissociation:** The extent to which an individual loses awareness of their surroundings
- **Emotional Involvement:** The level of emotional engagement with the activity.
- **Cognitive Involvement:** The mental effort and focus dedicated to the activity.

The IEQ uses a combination of Likert-scale questions to measure these dimensions, offering a detailed picture of how immersive an experience was (Drannalcox, 2024). To note that the score for the questionnaires is out a total of 7 points, 1 being the lowest score meaning less immersive and 7 being the highest score meaning more immersive.

4.1.1 Playtest 1 Results

Using the IEQ Scoring system, here are the results from each participant testing out the 3 initial prototypes:

Participant 1 - Brandon					
	Maze	Car	Boss		
	6	5	7		
	6	6	6		
	6	4	7		
	5	4	6		
	3	3	5		
	6	6	7		
	2	6	6		
	3	5	5		
	4	4	2		
	3	3	6		
	3	3	4		
	4	4	6		
	5	6	5		
	4	2	7		
	4	6	2		
	7	7	4		
	6	3	6		
	3	2	3		
	4	6	6		
	2	2	4		
	5	5	6		
	6	6	7		
Total	97	98	117		
Immersion x/10	6	6	9		
Maze Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
4	6	5	3	6	
4	2	3	3	6	
7	4	3	4	6	
6	5	4	2	5	
	3	6			
Total					
5.25	4.25	3.6	3.6	5.75	
Car Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
2	6	4	3	5	
6	6	3	2	6	
7	4	5	6	4	
3	6	4	2	5	
	3	6			
Total					
4.5	5.5	3.8	3.8	5	
Boss Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
7	7	6	6	7	
2	6	5	3	6	
4	6	5	6	7	
6	5	2	4	6	
	4	7			
Total					
4.75	6	4.4	5.2	6.5	
Participant 2 - Daniel					
	Maze	Car	Boss		
	6	4	7		
	7	6	7		
	7	4	6		
	2	1	4		
	1	1	4		
	7	2	7		
	5	4	7		
	5	5	5		
	2	1	5		
	6	1	7		
	4	3	4		
	6	3	7		
	4	1	6		
	5	2	6		
	3	3	6		
	6	2	7		
	5	1	4		
	2	1	4		
	6	3	7		
	3	1	5		
	6	4	7		
	7	3	6		
Total	105	56	128		
Immersion Q x/10	7	3	5		
Maze Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
5	7	2	6	6	
3	5	1	2	7	
6	6	5	6	7	
5	4	2	3	6	
	4	7			
Total					
4.75	5.5	2.8	4.8	6.5	
Car Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
2	2	1	1	4	
3	4	1	1	6	
3	3	5	3	4	
2	1	1	1	4	
	3	3			
Total					
2.5	2.5	2.2	1.8	4.5	
Boss Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
6	7	4	7	7	
6	7	4	4	7	
7	7	5	7	6	
4	6	5	5	7	
	4	6			
Total					
5.75	6.75	4.4	5.8	6.75	
Participant 3 - Hussein					
	Maze	Car	Boss		
	7	7	7		
	6	7	7		
	5	6	7		
	5	5	6		
	5	5	6		
	7	7	7		
	4	5	4		
	5	5	6		
	3	5	6		
	6	6	7		
	6	6	6		
	6	6	6		
	6	6	4		
	5	3	2		
	2	2	3		
	7	7	7		
	4	4	3		
	3	3	6		
	7	7	7		
	4	4	3		
	3	3	6		
	7	7	7		
	3	5	7		
	6	6	7		
	6	6	7		
Total	114	119	128		
Immersion x/10	10	10	10		
Maze Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
5	7	5	6	7	
2	4	5	3	6	
7	6	5	7	5	
4	6	3	3	6	
	6	6			
Total					
4.5	5.75	4.8	5	6	
Car Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
3	7	5	6	7	
2	5	5	3	7	
7	6	5	7	6	
4	6	5	5	6	
	6	6			
Total					
4	6	5.2	5.4	6.5	
Boss Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
2	7	6	7	7	
3	4	6	6	7	
7	6	6	7	7	
3	4	6	7	7	
	6	7			
Total					
3.75	5.25	6	6.8	7	
Participant 4 - Zyad					
	Maze	Car	Boss		
	5	4	7		
	7	5	7		
	7	5	7		
	6	4	3		
	6	5	5		
	7	6	6		
	4	3	5		
	5	3	5		
	3	3	2		
	5	5	6		
	6	5	6		
	5	5	6		
	6	4	5		
	5	5	6		
	7	4	2		
	6	4	4		
	2	4	4		
	3	2	3		
	7	4	6		
	5	3	6		
	6	4	6		
	5	4	6		
Total	117	91	113		
Immersion x/10	7	6	8		
Maze Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
5	7	6	5	5	
7	4	6	2	7	
6	5	5	7	7	
2	6	3	5	6	
	6	5			
Total					
5	5.5	5.2	4.8	6.25	
Car Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
5	6	4	5	4	
4	3	5	2	5	
4	5	3	4	5	
4	4	3	3	4	
	5	4			
Total					
4.25	4.5	4	3.6	4.5	
Boss Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
6	6	3	6	7	
2	5	5	3	7	
4	6	5	6	7	
4	5	2	6	6	
	6	6			
Total					
4	5.5	4.2	5.4	6.75	

Figure 36 - Playtest 1 Participant Scores (Viewable in [Appendix 2](#))

Maze Factor

Maze Factor Summary				
Challenge	Control	Real World Dissoc	Emot. Inv	Cogn. Inv
4.875	5.25	4.1	4.55	6.125

Figure 37 - Maze Score Average

The results suggest that for the Maze prototype, there was a high level of immersion among the participants. The highest scoring factor was the Cognitive Involvement scoring 6.125 showing that each participant was mentally engaged. The lowest scoring factor overall was the real-world dissociation scoring 4.1 followed by emotional involvement scoring 4.55 meaning that although they were aware of their surrounding environment, they weren't immersed deep enough into the game to lose track of their surroundings.

Challenge scoring 4.875 and Control scoring 5.25 shows that the participants found the prototype challenging but were in control of the player.

Car Factor

Car Factor Summary				
Challenge	Control	Real World Dissoc	Emot. Inv	Cogn. Inv
3.8125	4.625	3.8	3.65	5.125

Figure 38 - Car Score Average

The average results for the Car prototype suggest that there was a moderate level of immersion showing some variety between participants. Generally, Cognitive involvement scored the highest with a score of 5.125 showing that all the participants were attentive and engaged with this prototype. Second to that, Control scored 4.625 suggesting that players adapted to the controls of the vehicle effectively despite a few times some participants would spin out or crash into the surrounding environment. Challenge, Real-world Dissociation and Emotional Involvement scored below 4.0 indicating that the objective within the prototype was straight forward but not as immersive which makes sense as all the participants struggled when playing through the prototype. Although scores show some factors below 4.0, P3 did comment that they preferred playing the Car prototype over the Maze.

Boss Factor

Boss Factor Summary				
Challenge	Control	Real World Dissoc	Emot. Inv	Cogn. Inv
4.5625	5.875	4.75	5.8	6.75

Figure 39 - Boss Score Average

The results for the boss prototype show that this was a highly immersive experience. Again, the cognitive involvement scored the highest with 6.75 which does reflect on the performance of each participant trying to complete the objective for the prototype. Emotional Involvement and Control scored high with 5.8 and 5.875 respectively indicating that the participants had some emotional connection although some participants did have a bit of trouble working with the controls the first time, they played which did pose a challenge but got used to the control scheme quickly. Real-world dissociation had a moderate score of 4.75 representing that the participants did somewhat get lost within the game and lost awareness of the real-world surroundings. Lastly, challenge scored a 4.5 which does reflect with each participants performance as this prototype was the most challenging in terms of the objective but could be seen that with the slight greater difficulty reflects deeper immersion.

4.1.2 Playtest 1 Affinity Diagrams

Affinity diagrams were created to collectively display the notes written down whilst observing each participant play each prototype. Each diagram is split into 5 categories relating to controls, camera viewpoint, gameplay flow, visual challenges and initial first impressions.

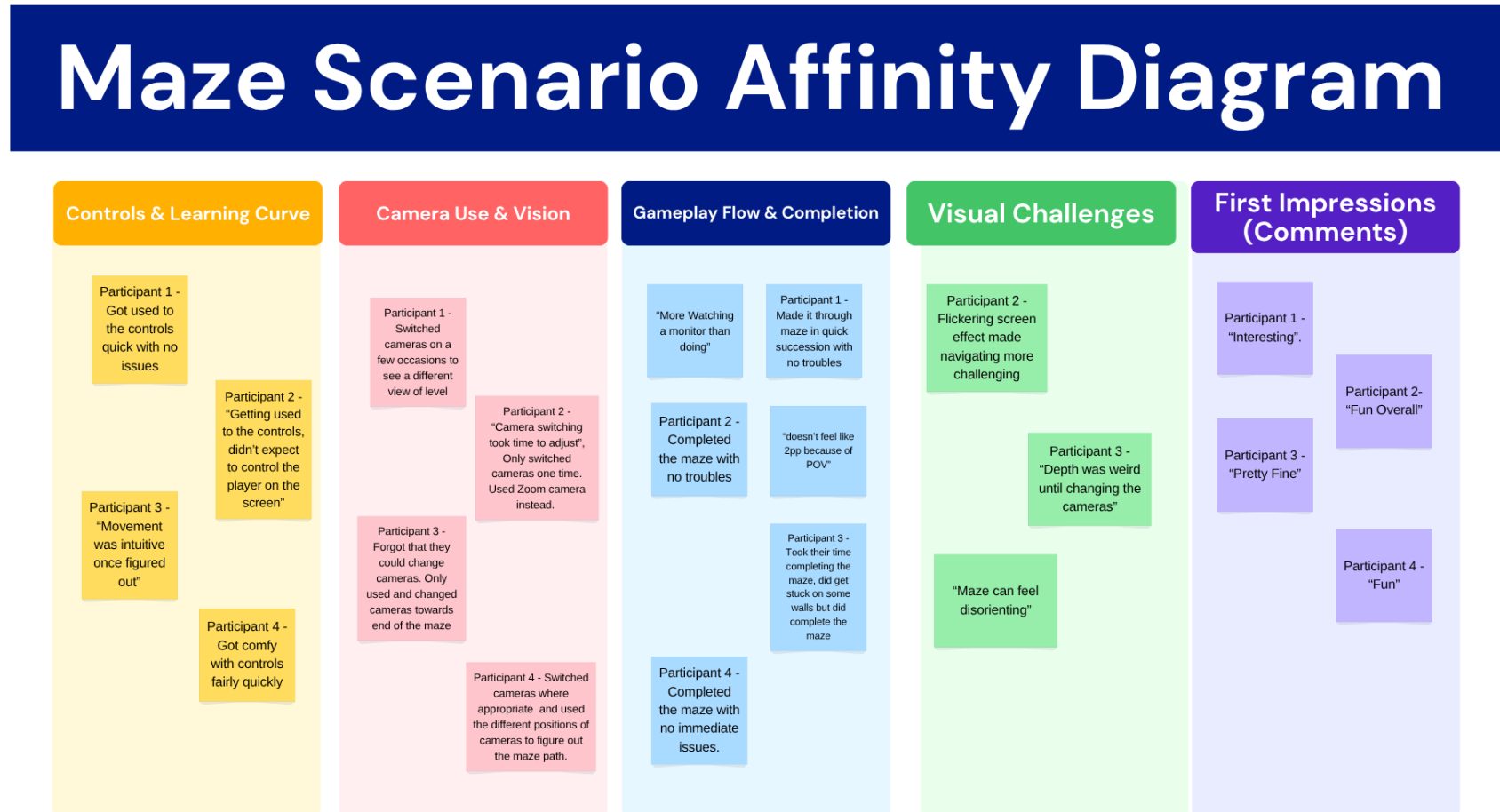


Figure 40 - Maze Affinity Diagram

Car Scenario Affinity Diagram

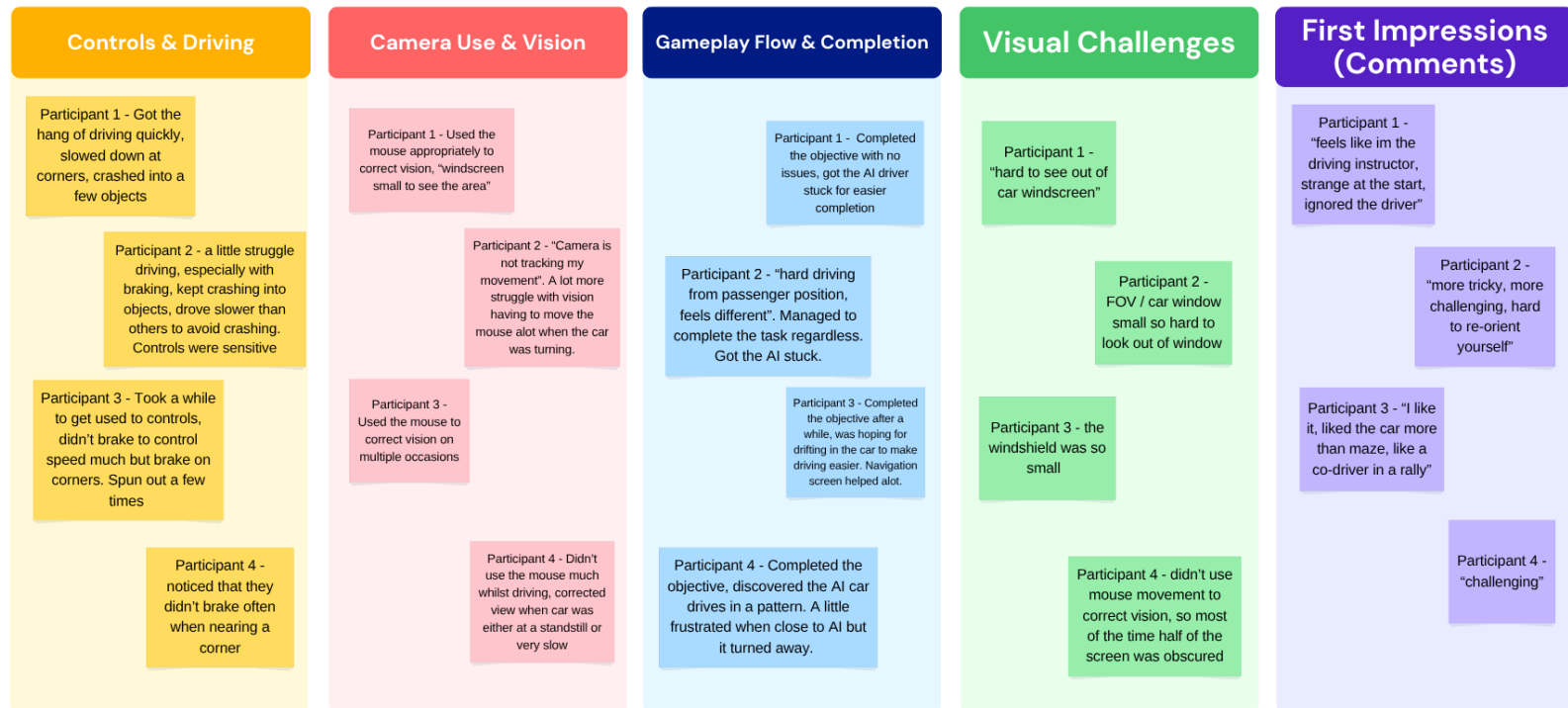


Figure 41 - Car Affinity Diagram

Boss Scenario Affinity Diagram

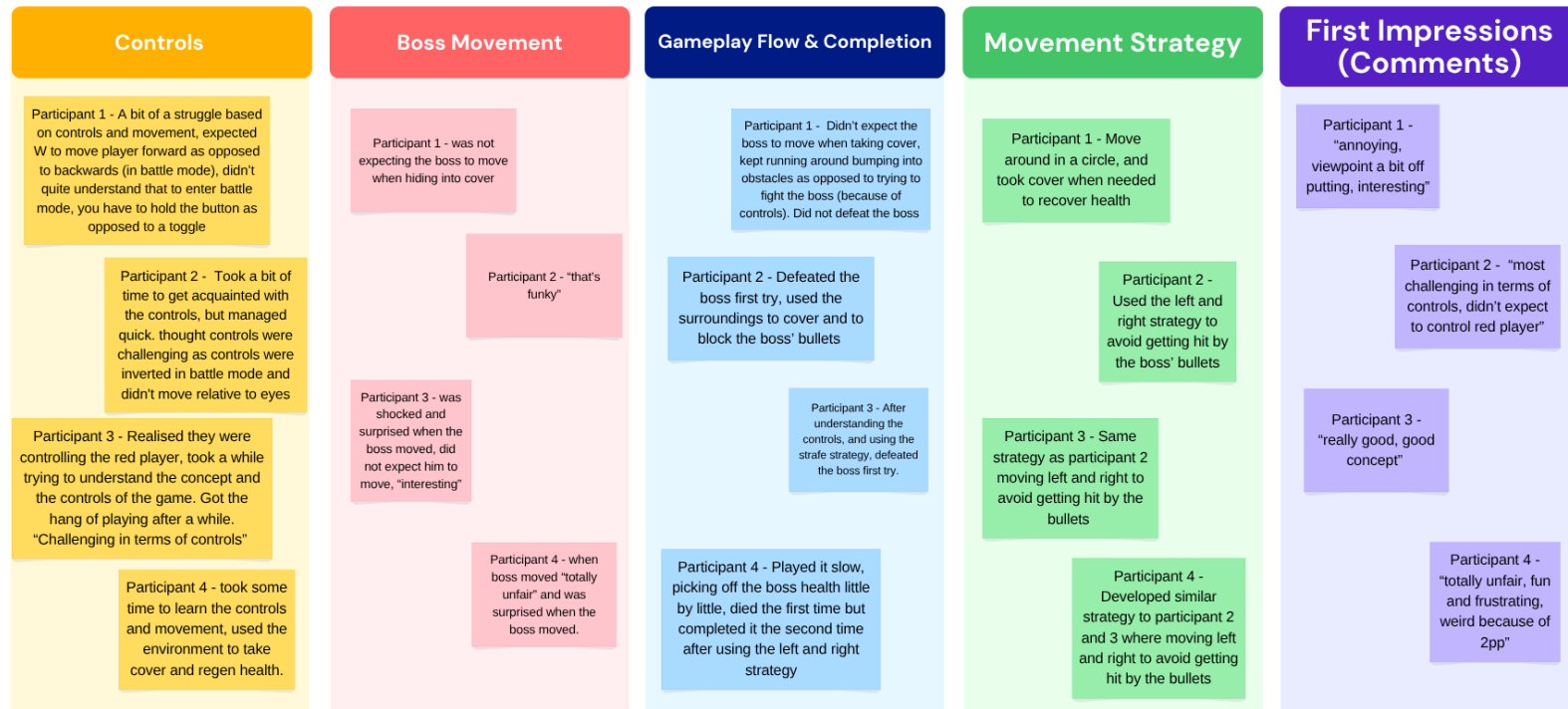


Figure 42 - Boss Affinity Diagram

4.1.3 Playtest 1 Data Gathering

During the playtest, a google form was used to collect additional data reflecting on the participants thoughts and to gather a better understanding as to what the second person perspective means to a more general audience. Google Forms and responses are available in [Appendix 2](#).

After playing each of the prototype scenarios, participants were asked some questions shown in **figure 43** describing what 2PP is in games which generated some interesting responses:

- P1 - “Unnatural and Confusing”
- P2 - “Playing from another person’s perspective”
- P3 - “Interesting, not used often, interesting in narrative”
- P4 - “Almost like first person, makes you think in terms of gameplay”

From what they’ve played, differentiating 2PP from the first-person perspective, responses were:

- P1 - “Camera is somewhere else”
- P2 - “Makes it harder, unlike 1PP, makes the controls harder because of the direction of movement (Boss Scenario) and the camera doesn’t move with the player”
- P3 - “Seeing your character in a nonlinear way, seeing them from a semi fixed position, watching them move around”
- P4 - “You feel separated, because you can see the first person, but you feel connected because you are controlling them. But the way your mind thinks on how to move doesn’t register how you want them to move”

Differentiating 2PP from a third-person perspective:

- P1 - “Camera on your opponent or other person”
- P2 - “More detached in 2PP, you view the character differently”
- P4 - “Much easier to control from 3PP, you feel a bit more attached to the game in 3PP. 2PP view you don’t feel connected to the character even though I can see them and control them”

Asking each participant which prototype demonstrated 2PP the best, 3 thought the Boss scenario and 1 thought the Car scenario demonstrated it the best. Based on participants responses:

Maze –

- P1 - “More watching a monitor than doing”
- P2 - “Doesn’t feel 2PP as much as Boss because of the view”
- P3 - “Can get disorienting”

Car –

- P1 - “Does feel like 2PP but afterwards you’re just driving, and you forget on the driver”
- P2 - “Least best demonstrated”
- P4 - “Physically aware you’re in the passenger seat, can see someone there and driving and he’s the one turning but I’m the one controlling them.”

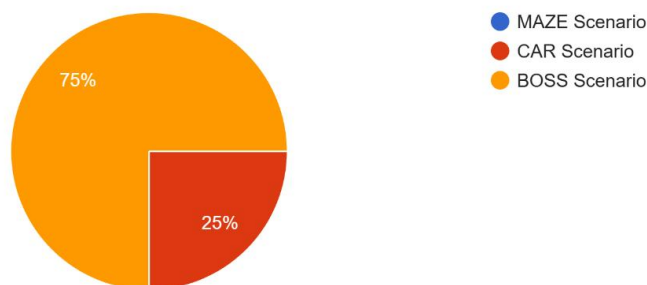
Boss –

- P1 - “Constantly reminding yourself you’re not the boss; you’re the one standing ahead”
- P2 - “Shooting at myself (At the boss)”
- P4 - “Boss was fun, the camera moved on its own and easier to guide the player”

Looking at all the data produced from the first playtest, the Boss prototype had the highest score of control, real world dissociation, emotional and cognitive involvement meaning participants were more immersed in playing the boss scenario compared to the others as it better demonstrated the second person perspective. Because the camera was fixed onto the boss, gazing at the player constantly whilst still having full control of the player made the scenario more engaging. The findings indicated that the perspective is strong when the viewpoint supports both the player’s actions and the objective whilst getting the feeling of being watched by someone.

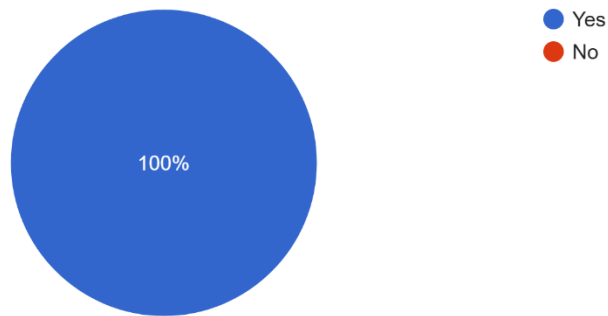
Which prototype scenario best demonstrated the second person perspective?

4 responses



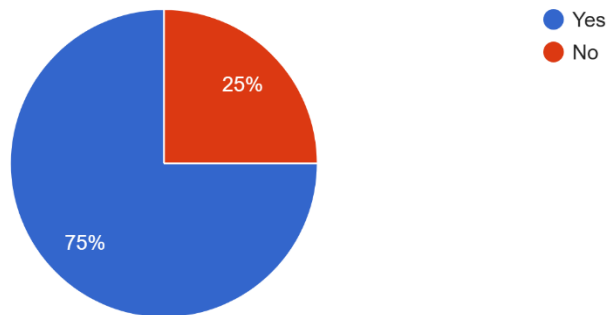
Could you imagine the second person perspective used effectively in other game genres?

4 responses



Would you be interested in playing a full game using only the second person perspective?

4 responses



Do you see the second person perspective as a novel gimmick or something has long term gameplay potential?

3 responses

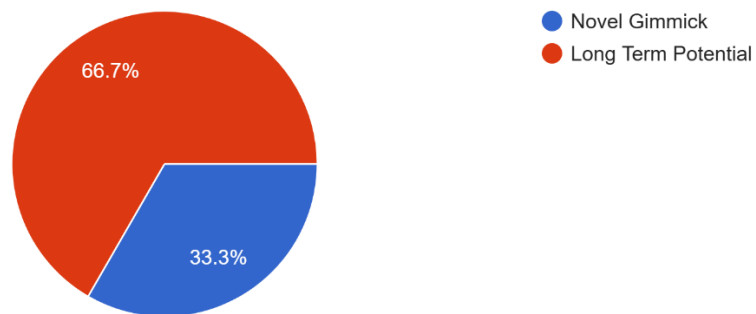


Figure 43 - Google Form 1 Responses

4.2 Playtest Session 2

The second playtest session was hosted during October 21st – 23rd with the help of 5 participants testing out the Journalist prototype which was created after the 3 initial prototypes. The google form had some additional questions added and some questions have been altered to suit the research based off this second prototype.

4.2.1 Playtest 2 Results

The second playtest session took place from 21st October – 23rd October. Here are the IEQ results for the Journalist prototype playtest session:

Participant 1 - Brandon (2)					
	Journalist				
	6				
	6				
	7				
	4				
	4				
	6				
	5				
	4				
	3				
	4				
	5				
	4				
	6				
	3				
	5				
	6				
	5				
	4				
	5				
	3				
	5				
	6				
	5				
	4				
	5				
	3				
	5				
	7				
Total	105				
Immersion Q	x/10	7			
Journalist Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
3	6	4	4	6	
5	5	4	5	6	
5	4	4	4	7	
6	3	3	5	3	
		5	5		
Total					
4.75	5.25	4	4.6	5.5	

Participant 2 - Abdullah					
	Journalist				
	6				
	7				
	6				
	3				
	5				
	7				
	7				
	6				
	5				
	6				
	5				
	5				
	6				
	2				
	6				
	7				
	4				
	7				
	4				
	7				
	7				
	7				
Total	123				
Immersion Q	x/10	8			
Journalist Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
2	7	3	6	6	
6	7	5	5	7	
7	5	6	7	6	
4	6	5	4	7	
		5	7		
Total					
4.75	6.25	4.8	5.8	6.5	

Participant 3 - Adam					
	Journalist				
	5				
	7				
	7				
	4				
	5				
	7				
	7				
	4				
	5				
	4				
	6				
	4				
	3				
	7				
	7				
	4				
	2				
	7				
	6				
	7				
	7				
Total	120				
Immersion Q	x/10	7			
Journalist Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
4	7	4	7	5	
3	7	5	2	7	
7	4	4	7	7	
4	6	5	6	7	
		5	7		
Total					
4.5	6	4.6	5.8	6.5	

Participant 4 - Michael					
	Journalist				
	6				
	6				
	6				
	4				
	4				
	3				
	4				
	5				
	5				
	2				
	5				
	7				
	7				
	4				
	3				
	5				
	2				
	5				
	6				
	5				
	5				
Total	100				
Immersion Q	x/10	5			
Journalist Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
5	3	4	5	6	
7	4	4	3	6	
7	5	4	5	6	
4	2	3	2	5	
		5	5		
Total					
5.75	3.5	4	4	5.75	

Participant 5 - Anita					
	Journalist				
	6				
	6				
	6				
	6				
	5				
	4				
	3				
	7				
	3				
	4				
	2				
	5				
	6				
	5				
	6				
	5				
	5				
	6				
	5				
Total	108				
Immersion Q	x/10	7			
Journalist Factors					
Challenge	Control	Real World	Emot. Inv.	Cogn. Inv.	
5	7	6	7	6	
5	5	5	5	6	
3	4	4	6	6	
4	2	3	5	6	
		3	5		
Total					
4.25	4.5	4.2	5.6	6	

Figure 44 - Playtest 2 Participant Scores

Journalist Factor

Journalist Factor Summary				
Challenge	Control	Real World Dissoc	Emot. Inv	Cogn. Inv
4.8	5.1	4.32	5.16	6.05

Figure 45 - Journalist Score Average

By averaging out the results from the 5 participants, this shows the journalist prototype had a moderately high level of immersion amongst the participants. Cognitive Involvement scored the highest of 6.05 meaning that the users were mentally engaged throughout the course of the prototype. Emotional Involvement and Control scored 5.16 & 5.1 respectively which suggests that the participants felt an emotional connection with the events going on within the game. The control being on the higher side of the scale indicates there was some difficulty in controlling the character and performing the actions. Challenge scored 4.8 meaning that the objective for the prototype seemed challenging for the participants and lastly Real-World dissociation scored 4.32 implies that although the participants were focused on the game, they weren't fully lost into the world and had some awareness of their surroundings.

4.2.2 Playtest 2 Affinity Diagram

This affinity diagram displays what was observed during the participants playing through the Journalist prototype. This diagram is split into 5 categories relating to controls, camera man, gameplay flow, visual challenges and initial first impressions.

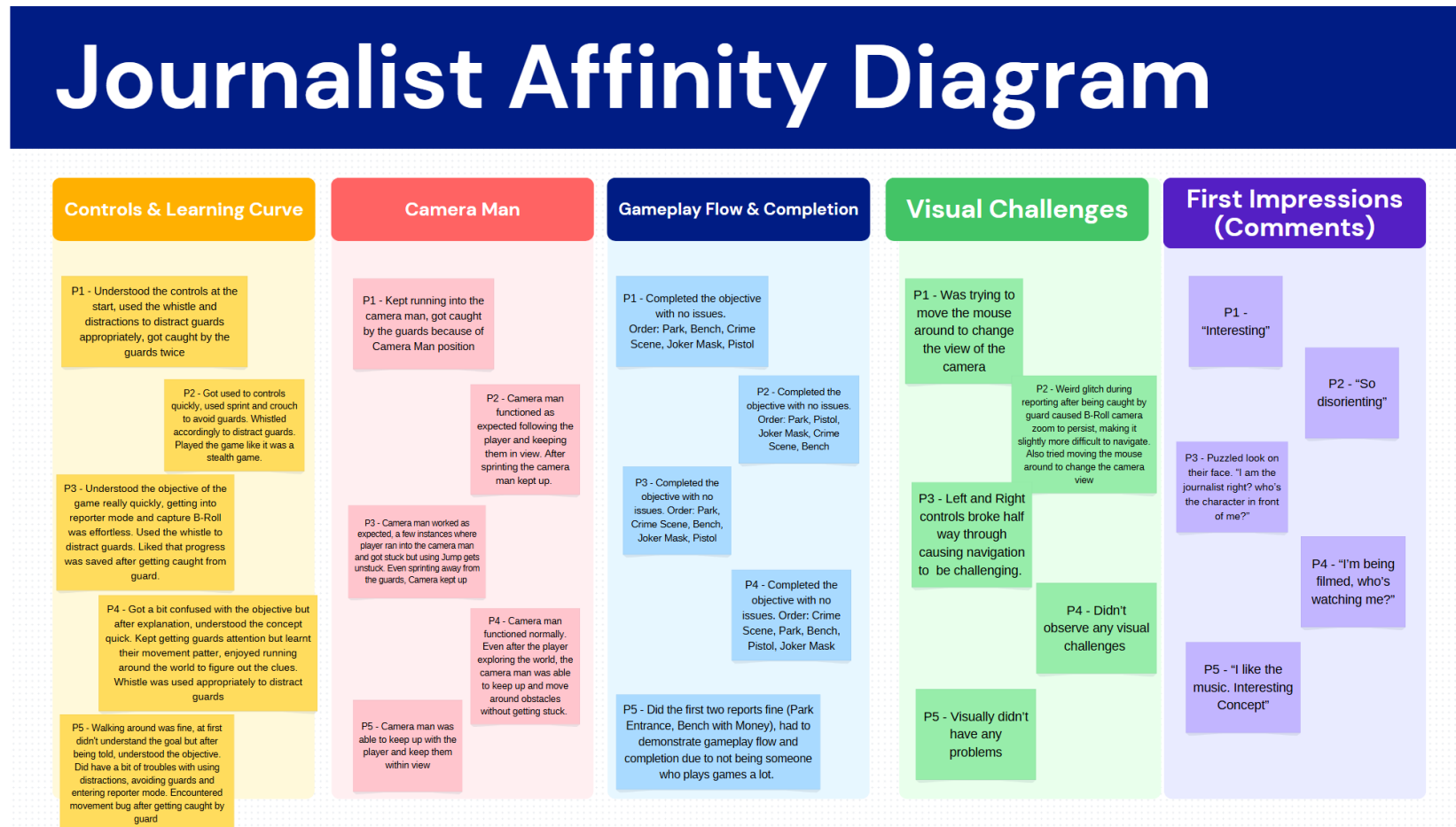


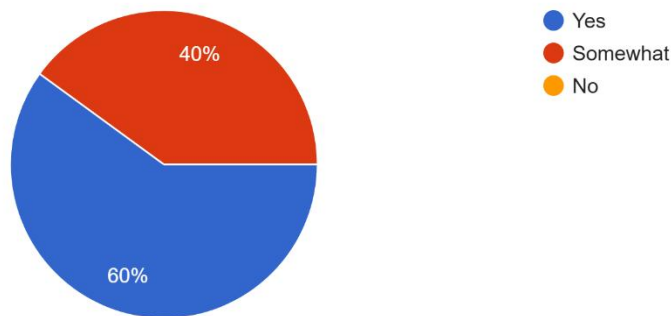
Figure 46 - Journalist Affinity Diagram

4.2.3 Playtest 2 Data Gathering

Same as the first playtest, the google form was used to gather some more information from each participant. For the journalist prototype, 3 people were able to identify 2PP being showcased but what was more interesting is that when asked who they were controlling, 1 person thought they were controlling the camera man and not the news reporter who is the main playable character in the prototype.

Were you able to identify how this prototype showcased the second person perspective?

5 responses



Who were you controlling? The news reporter or the camera man?

5 responses

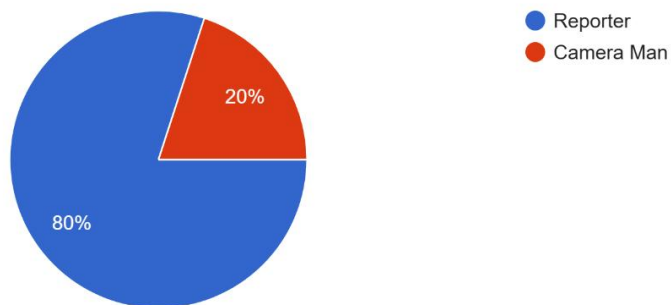


Figure 47 - Google Form 2 Responses

Here are the participants responses:

- P1 - “Different person holding the camera, following the player. Controlling both but following the news reporter but also playing as the news reporter”
- P2 - “No control of the camera, only in control of the movements of the main player character. The NPC is still an NPC; I have no control over his position and so it feels detached from me”

- P2 - “Observing the player’s movements, like watching the person viewing them from every direction.”
- P3 - “I was controlling the news reporter but watching from the perspective of the camera man.”
- P3 - “There’s a disconnect between the view and the movement while going around the world”
- P4 - “Giving an input, the reporter moves, the camera man follows the reporter but not the path that reporter takes.”
- P5 - “No control of the camera, only control of the player.”

Was it clear that you were seeing the news reporter from the perspective of the camera man?

5 responses

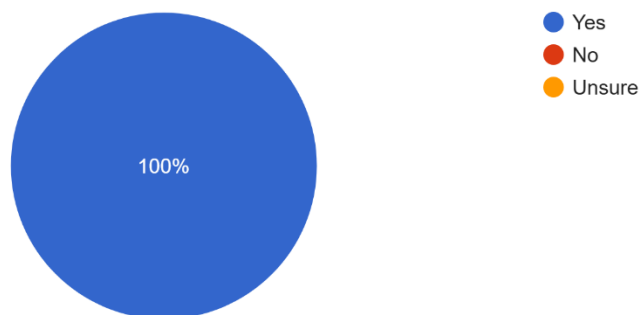


Figure 48 - Google Form 2 Response

To emphasize 2PP in the Journalist prototype, the idea is that you are seeing yourself (the news reporter) from the perspective of the Camera Man. Responses from participants follows:

- P1 - “After a while, it gets clearer”
- P2 - “The game made it clear when you record the footage, he’s looking at the camera, so I know ‘that’s the camera man’”
- P3 - “Intuitive, you see the UI and the reporter information on the screen”
- P4 - “You see the news reporter from the Camera man’s perspective and the phone camera is held by the camera man”
- P5 - “Only able to see the news reporter, recording them on the phone”

Describing 2PP in games, gathered some more insight about the perspective as a whole:

- P1 - “Another person, separate person but controlling the player”
- P2 - “Second person is controlling a character but watching them from another perspective. A blend of third person but breaks the first-person barrier”
- P3 - “Separation between the game and real world, watching a recording rather than playing”

- P4 - “Feels like you’re in control and not in control at the same time. Not in control of the viewpoint”

What differentiates 2PP from the first person was asked and responses show:

- P1 - “Seeing the player for what they are and what they’re doing”
- P2 - “First person, the camera is in their eyes. Second person you’re watching them from the outside”
- P3 - “The world we live in is in first person, wouldn’t imagine looking at myself while doing a certain task”
- P4 - “The restriction of the viewpoint of the main character, cannot control the projection of the main character. You can see the player but in first person you can’t see yourself”

And for third person:

- P1 - “Using B-Roll to zoom in to look elsewhere and not just the player”
- P2 - “Second person you’re controlling their movement and watching, you’re not that character but you’re controlling them. Third person you’re the character but looking at them from outside their body, the camera follows the player”
- P3 - “The view is closer to third person although it can be said that a typical 3PP game is like this but with the camera lagging.
- P4 - “Cannot control the camera of the player”

Would you be interested in playing a full game using only the second person perspective?

5 responses

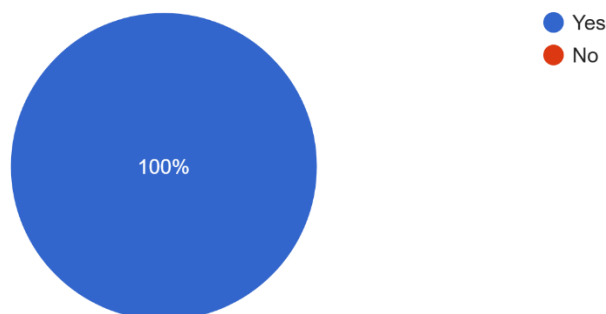


Figure 49 - Google Form 2 Response

Again, all the participants would be interesting in playing a full game that was only using the second person perspective.

When seeing if 2PP is a novel gimmick or having long term potential, there were some mixed responses. One said that it's not a gimmick but can't see it in a full game but having certain scenarios within that game could benefit with the alternate perspective, but another person said that it's currently a gimmick as you'd need to develop a story and control scheme around the second person.

Do you see the second person perspective as a novel gimmick or something has long term gameplay potential?

5 responses

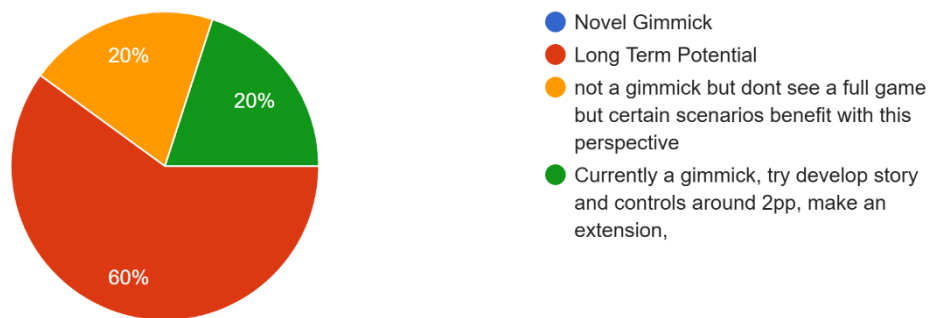


Figure 50 - Google Form 2 Response

Overall, the results from the second playtest shows that the moderately high immersion scores are mostly because of the novel viewpoint of the journalist prototype. Because the camera moved independently from the player, it created some confusion as to who they were controlling, creating that disconnect. But in terms of immersion based on the scores, was above average yet the cognitive involvement remained the highest score due to the constant reassurance that the camera is not being controlled, the player is the one being controlled. Altogether, the concept of the second person used in the journalist prototype was effective however there's still room for improvement such as clearer visual cues and mechanics to help players understand the premise of the game world.

4.3 Prototype Output

Throughout the process of this project, 4 mini games have been produced that showcases the second person perspective in a wide multitude of genres. The Maze covers puzzle, Car covers racing and driving, Boss covers shooters and Journalist covers action/adventure.

4.3.1 Maze Output

The Maze is the simplest prototype created. The main environment is a security office observing a computer that shows a Maze and the player. The second person POV is observing the red capsule player from the view of the security officer. As the gameplay was simple, no bugs were discovered during the development and playtest.

4.3.2 Car Output

The Car prototype featured a self-driving AI car that drives around following a set path in the level. The environment is within a city scape. The drivable car features two people, the driver and the passenger. The world is seen through the perspective of the passenger whilst the player can control the driver and drive the car around. For bugs, a few were encountered but didn't hinder gameplay, the AI driver sometimes got stuck on the geometry of the world or would struggle to drive back to the waypoints. The drivable car was difficult to control at top speeds and sometimes would cause the car to spin out when attempting to take turns.

4.3.3 Boss Output

The Boss prototype delivered the most promise with showcasing the second person perspective more effectively than the other prototypes within playtesting. The game features a small boss arena with a few obstacles across the level for the player to avoid, run around, or hide behind to avoid the Boss' sight. If the player hides for too long, then the Boss jumps and moves between different platforms. The world is seen through the Boss' eyes that's constantly tracking and looking at the character which the player controls. For bugs, there was one bug where sometimes the bullet that the Boss would shoot would not register hitting the player. One oversight is with the vision, by positioning yourself behind one of the pillars, the boss would be able to see the player and shoot but not hit the player, however the player can shoot back at the boss and easily defeat them.

4.3.4 Journalist Output

The journalist prototype is by far the most detailed in terms of gameplay, environment, controls and ambience with dialogue, AI enemies with their own states, the camera man following the player and a linear mini story for the player to follow. Using the same City landscape as the Car prototype but isolating the play area to just the park, features multiple areas for the player to report and film clues and evidence, and having guards roaming around provided challenge and created situations where stealth would be ideal. Using the knowledge learned from the 3 other prototypes, 2PP is best showcased within the journalist prototype since you are controlling the news reporter, but you are seeing yourself through the perspective of the Camera man who is holding the mobile phone. There were quite a lot of bugs present such as when entering reporter mode and being caught by the guard, sometimes the controls would invert by itself making it harder to run around and move the player. When using one of the distractions placed, if the player ran outside the guard zone, the guard would disregard the distraction and go back into the Patrol state, this would also happen with the whistle distraction, however if you whistle outside of the guard zone, the guard will constantly detect you even if you're outside of the zone. When in reporter mode, sometimes when exiting the mode, the prompt text would appear, and the controls would stop working due to the way it was programmed.

4.3.5 Playtest Session Images

Recordings of Playtest sessions are available in [Appendix 4](#).



Figure 51 - Phase 2 Prototype Main Menu



Figure 52 - Maze Prototype Gameplay



Figure 53 - Car Scenario Gameplay



Figure 54 - Boss Scenario Gameplay







Figure 55 - Journalist Prototype Screenshots

5 Discussion

Discussing what has been achieved throughout the process of this project, I believe that the objectives set earlier within this report have been achieved successfully.

- 1. To analyse the second person perspective within existing video games and interactive narratives** – This Objective was partially successfully due to not being able to explore the narrative side of the perspective. However, being able to identify the already existing games that present the second person perspective in their own unique way was interesting to see how the perspective was implemented did aid in creating the prototypes that I've built. *Driver: San Francisco* and *Psychonauts* provided the most help with understanding how to implement the second person by situating the camera inside another body whilst trying to always keep the main playable character in the centre frame. To add more emphasis on second person, having the arms of the non-playable character sells the illusion of second person providing that weird disconnect, separation and unique view of the game.
- 2. Explore how the second person perspective impacts the player's immersion** – This objective was strongly met as 4 different scenarios were created ranging from different genres on how the second person can be implemented: Puzzle, Driving, Shooter and Action/Adventure. The alternate perspective had quite a lot of impact on immersion during the playtesting, where the *Boss* and *Journalist* prototype had most of the immersion scores being above 7/10 whereas the perspective showcased within the *Maze* and *Car* prototype disrupted immersion having scores between 5 - 6/10. Overall, the participants displayed high levels of immersion when playing each prototype.
- 3. To identify the potential gameplay experiences that are unique to the second person perspective** – This objective was partially met with the creation of the *Journalist* prototype. During testing, there were responses that the prototype participants played said it was a game that they haven't seen or played before. Although the methods and gameplay elements used within the *Journalist* prototype have been done before in some games using first and third person, the use of second person can open up a lot of gameplay possibilities that utilise the camera man being an independent entity, an example being the camera man can see something that the player wouldn't have been able to see because of the alternate perspective that one person can see at any given time. Novel mechanics weren't made during this project although with enough time, skill and thoughts, new different mechanics can be produced to work well with the second person perspective.

4. Compare the second person perspective with first and third person perspectives in relation to storytelling – This objective wasn't met due to the time availability. With enough time and effort, each of the prototypes created for this project could've integrated a first- and third-person camera view as well as a story to go along with each objective to see if the second person does affect the game in relation to storytelling. However, during the Journalist playtest, this specific objective was asked to the participants to gather an insight as to what their thoughts were. A couple responses included:

- “It would make it more immersive especially in the context of the Reporter and Camera Man because it's like you're watching a news report unfold live, whereas in first person you can't see the product. For example, even though I'm controlling the reporter, as the camera man I can see the news reel whereas in first person, If I was the reporter I'd be looking and speaking at the camera man. It wouldn't feel like I'm in the game. Second person delivers the idea of being a news reporter.”
- “The big difference on how it affects storytelling is that you're looking at a story. You as the player is looking at the events that is beyond your control, i.e. it's external from you, you cannot control the events in second person. First person makes you feel that you are the player, you have the choice and free will within the story. Third person is the middle ground of both, you are in control but different from the player character, you're the puppeteer of the character. In second person, I would see the character making its own decision, I am not the puppeteer, in third person I can see the other angles and first person makes me feel I am the puppet, I have free will. With engagement, I am engaged as to what the character is doing, third person is what can this character do? First person is what can I do?”

From the participants responses after playing the Journalist prototype, there is a much larger potential for telling a story within the second person perspective which could be more engaging and exiting than playing the same story told within first or third person.

5. Evaluate player's perceptions of the second person perspective through user testing –

This objective was strongly met as participants were required to fill in the IEQ survey after playing each prototype. Although many people showed interest in participating, because the testing was only possibly in person, a total of 9 participants were able to be involved in testing out the prototypes created for this project which was highly appreciated. With more time, online testing would've allowed many more participants to gather data for this project. The data collected showed that the highest scoring factor during testing was the Cognitive involvement meaning that participants were mentally engaged with the second person perspective. Real-World Dissociation was often the least scoring factor meaning that although they were engaged and immersed within the game, it wasn't immersive enough for them to forget about the real world and their surroundings. Challenge, Control and Emotional Involvement were in the middle meaning that there was a decent amount of difficulty when completing the objectives and controlling the player on screen. Emotional Involvement meant that the participants were connected to the player and their interactions.

6 Evaluation, Reflection & Conclusion

6.1 Evaluation

I had a Gantt chart created to showcase the work plan and structure that I was going to follow throughout the duration of this project. For the most part, the work plan was followed however there were a few additions such as the extra development of a 4th prototype and the second set of playtesting.

Before finalising the current objectives for this project, my previous objectives and questions for this project were:

Research Questions

- How does the 2nd person perspective influence player immersion during gameplay?
- What gameplay opportunities do 2nd person games offer compared to traditional 3rd and 1st person games?
- What impact does 2nd person perspective have on story telling within a gameplay experience?
- Does the 2nd person perspective offer unique gameplay experiences compared to traditional 1st and 3rd person perspectives?
- Are there creative options that offer new gameplay experiences that haven't been explored yet or is the minimal 2nd person perspective a novelty experience?

Research Objectives

- Analyse the Second Person Perspective within existing video games and interactive narratives.
- Compare the Second Person Perspective with First and Third Person perspectives on storytelling within a gameplay experience.
- Explore how the Second Person Perspective impacts the player's immersion.
- Identify the potential gameplay experiences that are unique to the second person perspective.

Later, the objectives and questions were combined to create the 5 objectives that I've discussed within this report.

The literature examined was extremely helpful as it showed the possibilities of the second person that's already been implemented in games. If games such as Driver: San Francisco with the level pursuing the player through the NPC's eyes and Psychonauts' Den Mother boss fight seeing yourself through her vision didn't feature such prominent, unique gameplay and perspective within those experiences, then researching and understanding what the second person meant in games would've been difficult and would have required a lot more research than just looking at games.

As I've had prior experience using Unity and familiar with the engine's interface, it allowed me to play around with the idea of the second person by experimenting with the different possibilities the cameras and game objects could be positioned within the 3D space.

To evaluate on the overall process for this project, I believe that I was able to succeed in exploring the second person perspective in a wide variety of genres through the 4 prototypes that I've created. My thoughts about the second person in games is that you are seeing yourself (the player) from the point of view of another person or entity. A person being like the camera man in the journalist prototype, and an entity being a security camera or a television showing the player on screen.

6.2 Reflection

To reflect on the project, I did enjoy exploring the different possibilities of the second person perspective in games and was excited to see what I could've created with my currently ability and skillset. Overall, I am proud with what I have created especially being able to develop 4 playable prototypes that showcase the second person in their own unique way and in a different genre all together. What sparked joy was when people play tested the prototype and gathering their feedback and insights about the perspective, being able to introduce this alternative perspective to gamers with a common point of view to a new medium. Although so much has been created for this project, because of the word constraint, there are a lot of things that cannot be detailed and discussed within this report.

Reflecting on each prototype:

Maze

The concept for the maze was originally a headless player that would have puzzle-like chambers like Portal, however when I was developing the maze, it started off as just cameras in the corner for the player to look and cycle through. Creatively, this lacked the imagination that I wanted to show initially, and it ended up being the least pleasurable prototype.

Car

The car prototype started off as just driving around an assault course with cones directing the player, however, when messing around with a secondary independent vehicle that drove by itself, it eventually turned into a chase game. The idea was to have the player chase down the AI driver and crash into it, taking inspiration to missions from The Simpsons: Hit and Run where you crash into other characters cars and blow them up. Halfway through I wanted to implement a second alternative camera that was attached to the AI car that faced towards the player vehicle to emphasise the second person showing yourself from the perspective of the car you're chasing but I didn't have time to add that feature. One thing that I wish I could change but I couldn't due to the asset pack was the vehicle. I used a sports vehicle from a prototype asset pack which worked well with Unity's vehicle colliders, however when trying to use a different vehicle the car's suspension was too bouncy and would rock the car left and right, making driving unappealing.

Boss

The boss prototype started off slow as I did not have the proper direction I wanted to take when developing. Firstly, it started with creating some platforms for the arena and go on from there. For this one, I used Unity's Cinemachine cameras for experimentation as it had lots of features such as camera tracking, splines and an easy way to add a head bob to provide a little life to the boss. Currently in the game, the boss glides across the arena to 3 determined positions on the map which does work, however I wanted to make the boss

have a jumping motion. One feature I would've wanted to implement was night vision goggles or an overlay on the camera to really display that you are seeing everything from the Boss's view.

Journalist

Lastly, the Journalist was the most ambitious yet buggiest prototype that I've developed. Before this prototype, I was initially going to develop a first-person heist game with a narrator explaining things that you've done. For example, "You walked into the bank, you took the keycard, you stole the money" etc, but I was struggling to implement the things I wanted, and it didn't seem "second person". After making a google form and asking people online for ideas, Road 96 was mentioned and after looking at screenshots, a journalist type game was thought of. Visually, this prototype is the most polished and more appealing than the other prototypes as I used an animation asset pack which had a playable locomotive player, however transitioning the already existing prefab into second person was the most challenging. The camera dealt with the animation, the lock on feature and the controls. There was a lot of trial and error to get what I wanted to work but there are bugs such as the controls inverting or not working at all. Ideally it would've been better to make my own script and player but due to time constraints and skill limitations, using the asset pack sufficed.

6.2.1 Further work

A lot of the features I wanted to implement into each of the prototypes were optimistic, but after developing the journalist prototype, I believe that the idea, concept and the unique view of the second person could be useful for future researchers and developers to expand on the idea of the perspective, and create their own games using this paper as a guideline of what goes into a Second Person Game. For the Journalist game, in the future it can be developed into a full-fledged game with a more robust reporting mechanic that grades you for the quality of the recording and report, and shows the recorded reports in the sequence that you reported in, allowing you to uncover secrets and create a story for a news blog. One feature that could be implemented is incorporating the car prototype by having the news reporter and the camera man drive around the country to find areas to report on, or to have a mini sequence of the camera man recording the reporter as he drives explaining the story in a mini cutscene. With enough time, effort and resources I think this could introduce a whole new genre and vision for gamers to explore and enjoy and to try something new that's outside of the already existing first and third person games that may not be able to capture the same immersive experience as the second person.

6.3 Conclusion

To conclude this report, there is still a gap in games regarding perspective where people are looking for something new and exciting, and if more games were created using the second person perspective, it would add and enhance to the current gaming community. As for what I've researched, I believe that I have been able to showcase exploring what games would look like if they were in the second person perspective and I am happy with the decisions made and have approached undertaking this feat in developing the prototypes for the beneficiaries to reflect and apply their own ideas with the help of this report.

References

- Alance AB (2025) *Maze Generator*, <https://www.mazegenerator.net/>.
- Bomb, G. (2011) ‘Second Person Shooter Zato’, <https://www.giantbomb.com/second-person-shooter-zato/3030-44604/> [Preprint].
- Chollet, F. (2022) *2nd-person game explanation*, <https://x.com/fchollet/status/1483076497413836800?lang=en>.
- Denisova, A. and Cairns, P. (2015) ‘First person vs. Third person perspective in digital games: Do player preferences affect immersion?’, in *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery, pp. 145–148. Available at: <https://doi.org/10.1145/2702123.2702256>.
- DraKulis (2020) *South Vietnam 1968 (Vietnam Flashbacks) Call of Duty Black Ops Cold War - Part 12 - 4K*, <https://www.youtube.com/watch?v=MTWINKIhmZg>.
- Drannalcox (2024) *Measuring Immersion: The IEQ Explained*, <https://www.eworkresearch.org/measuring-immersion-the-ieq-explained/>.
- How to play guide for Road 96 | Road 96 Wiki | Fandom* (no date). Available at: https://road96.fandom.com/wiki/How_to_play_guide_for_Road_96 (Accessed: 30 October 2025).
- Marino (2019) *Screencheat*, <https://www.giantbomb.com/screencheat/3030-47190/>. Available at: <https://www.giantbomb.com/screencheat/3030-47190/> (Accessed: 22 August 2025).
- MasterClass (2021a) *Complete Guide to Different Types of Point of View: Examples of Point of View in Writing - 2025 - MasterClass*. Available at: <https://www.masterclass.com/articles/complete-guide-to-point-of-view-in-writing-definitions-and-examples> (Accessed: 22 August 2025).
- MasterClass (2021b) *What Is Second Person Point of View in Writing? How to Write in Second Person Narrative Voice With Examples*, <https://www.masterclass.com/articles/what-is-second-person-point-of-view-in-writing-how-to-write-in-second-person-narrative-voice-with-examples>.
- Merriam-Webster (no date) *First, Second and Third Person Explained*, <https://www.merriam-webster.com/grammar/point-of-view-first-second-third-person-difference#:~:text=Second%20Person,-Second%20person%20narration&text=The%20reader%20is%20immersed%20into,novel%20Bright%20Lights%20Big%20City>.
- Pauen, M. (2012) ‘The Second-Person Perspective’, *Inquiry*, 55(1), pp. 33–49. Available at: <https://doi.org/10.1080/0020174X.2012.643623>.
- Robinson, N. (2019) *This Is What a ‘Second-Person’ Video Game Would Look Like*, https://www.youtube.com/watch?v=mC8QoRa8y_Q. Available at: https://www.youtube.com/watch?v=mC8QoRa8y_Q (Accessed: 22 August 2025).
- SteamDB (2025a) ‘Steam Game Releases Summary for First Person’, <https://steamdb.info/stats/releases/?tagid=3839> [Preprint].

SteamDB (2025b) ‘Steam Game Releases Summary for Third Person’,
<https://steamdb.info/stats/releases/?tagid=1697> [Preprint].

Unity (2025) *Create a car with Wheel colliders*,
<https://docs.unity3d.com/6000.2/Documentation/Manual/WheelColliderTutorial.html>.

Zhou, J. (2022) ‘Where Is My Head?: Explorations of Second-Person Perspectives Used in VR Games’,
pp. 4–31.

Appendices

For access to any of the OneDrive links, please contact my supervisor (Stuart Scott)

Appendix 1 – GitHub Copilot AI

[GitHub Copilot AI Prompts and Generations Video](#) (21:22 Minutes)

Appendix 2 – Playtest Documents

All the Playtest documents including Google Forms, IEQ Survey, Signed Consent forms and Information Form

[Playtest Documents](#) (5mb)

Appendix 3 – Playable Prototypes

[Maze, Car & Boss](#) (WebGL Build)

[Journalist](#) (WebGL Build)

[Maze, Car & Boss](#) (Executable 178MB)

[Journalist](#) (Executable 107MB)

Appendix 4 – Playtest Recordings

All the playtest recordings from both sessions

[Playtest Recordings.zip](#) (14.6gb)

Appendix 5 – Unity Project

This is the Unity Project containing all assets, scenes and scripts

[Unity Project](#) (6.28gb)

Appendix 6 – INM363 Submission Files

All submission files

[Ali Al-Dibouni INM363 Submission Files](#) (22.2gb)